

Geometric Inequalities

23, 30 November, 14 December 2019

Contents

1 Notations 2

2 Some Cubic Equations related to Elements of a Triangle 2

2.1 Cubic Equation for Sides 2

2.2 Cubic Equation for Reciprocals of Sides 3

2.3 Cubic Equation for Altitudes 3

2.4 Cubic Equation for Reciprocals of Altitudes 3

2.5 Cubic Equation for $s - a, s - b, s - c$ 3

2.6 Cubic Equation for Reciprocals of $s - a, s - b, s - c$ 4

2.7 Cubic Equation for Exradii 4

2.8 Cubic Equation for Reciprocals of Exradii 4

3 Some Geometric Inequalities 5

3.1 Trigonometric Inequalities 5

3.2 Inequalities Involving Elements of a Triangle 6

3.3 Erdős-Mordell Inequality and Some Related Inequalities 13

3.4 Miscellaneous Geometric Inequalities 21

1 Notations

For any triangle ABC , we use the following notations:

- 1.1. $a = BC, b = CA, c = AB$;
- 1.2. $A = \angle BAC, B = \angle CBA, C = \angle ACB$;
- 1.3. M_a, M_b, M_c are the midpoints of BC, CA, AB , respectively;
- 1.4. m_a, m_b, m_c are the lengths of the medians from A, B, C to BC, CA, AB , respectively;
- 1.5. h_a, h_b, h_c are the lengths of the heights from A, B, C to BC, CA, AB , respectively;
- 1.6. w_a, w_b, w_c are the lengths of the internal angle bisectors to the triangle of $\angle BAC, \angle CBA, \angle ACB$, respectively;
- 1.7. v_a, v_b, v_c are the lengths of the internal angle bisectors to the circumcircle of $\angle BAC, \angle CBA, \angle ACB$, respectively;
- 1.8. k_a, k_b, k_c are the lengths of the symmedians from A, B, C to BC, CA, AB , respectively;
- 1.9. R is the circumradius, r is the inradius, r_a, r_b, r_c are the exradii;
- 1.10. r_a, r_b, r_c are the exradii of the excircles opposite A, B, C , respectively;
- 1.11. p_a, p_b, p_c are the distances from the circumcentre to BC, CA, AB , respectively;
- 1.12. R_1, R_2, R_3 are the lengths of AP, BP, CP , respectively, where P is an arbitrary point inside $\triangle ABC$;
- 1.13. r_1, r_2, r_3 are the distances from the point P to BC, CA, AB , respectively, where P is an arbitrary point inside $\triangle ABC$;
- 1.14. P_a, P_b, P_c are the feet of the perpendiculars from P to BC, CA, AB , respectively, where P is an arbitrary point inside $\triangle ABC$;
- 1.15. Δ is the area of $\triangle ABC$;
- 1.16. $[P_1 P_2 \cdots P_n]$ is the area of the (convex) polygon $P_1 P_2 \cdots P_n$, where $n \geq 3$;
- 1.17. s is the semi-perimeter of $\triangle ABC$;
- 1.18. O is the circumcentre, H is the orthocentre, I is the incentre, G is the centroid, N is the Nine-point centre, K is the symmedian point, N_a is the Nagel point, G_e is the Gergonne point.
- 1.19. K_a, K_b, K_c are the points where the symmedians from A, B, C intersect BC, CA, AB , respectively;
- 1.20. N_a, N_b, N_c are the points where the cevians from A, B, C passing through N_a intersect BC, CA, AB , respectively;
- 1.21. G_a, G_b, G_c are the points where the cevians from A, B, C passing through G_e intersect BC, CA, AB , respectively;

2 Some Cubic Equations related to Elements of a Triangle

2.1 Cubic Equation for Sides

We start with

$$a = 2R \sin A = 4R \sin \frac{A}{2} \cos \frac{A}{2}, \quad \text{and} \quad s - a = r \cot \frac{A}{2}.$$

Then

$$1 = \cos^2 \frac{A}{2} + \sin^2 \frac{A}{2} = \frac{a(s-a)}{4Rr} + \frac{ar}{4R(s-a)}$$

and so

$$a^3 - 2sa^2 + (r^2 + s^2 + 4Rr)a - 4Rrs = 0.$$

Therefore we see that a is a root of the cubic equation in x given by

$$x^3 - 2sx^2 + (r^2 + s^2 + 4Rr)x - 4Rrs = 0. \tag{1}$$

Similarly, b and c are roots of (1).

Consequently, we have

- 2.1. $a + b + c = 2s$,
- 2.2. $ab + bc + ca = r^2 + s^2 + 4Rr$,
- 2.3. $abc = 4Rrs$.

2.2 Cubic Equation for Reciprocals of Sides

We use (1) to obtain the cubic equation for the reciprocals of sides, namely $\frac{1}{a}$, $\frac{1}{b}$ and $\frac{1}{c}$ are the roots of the cubic equation in x given by

$$4Rrsx^3 - (r^2 + s^2 + 4Rr)x^2 + 2sx - 1 = 0. \quad (2)$$

Consequently, we have

$$2.4. \quad \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{r^2 + s^2 + 4Rr}{4Rrs}$$

$$2.5. \quad \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} = \frac{1}{2Rr}.$$

2.3 Cubic Equation for Altitudes

We note that from (2) we have

$$4Rrs\left(\frac{1}{a}\right)^3 - (r^2 + s^2 + 4Rr)\left(\frac{1}{a}\right)^2 + 2s\left(\frac{1}{a}\right) - 1 = 0.$$

Then we use the fact that $\frac{1}{a} = \frac{h_a}{2rs}$ to obtain

$$2Rh_a^3 - (r^2 + s^2 + 4Rr)h_a^2 + 4rs^2h_a - 4r^2s^2 = 0$$

and we see that h_a is a root of the cubic equation in x given by

$$2Rx^3 - (r^2 + s^2 + 4Rr)x^2 + 4rs^2x - 4r^2s^2 = 0. \quad (3)$$

Similarly, h_b and h_c are roots of (3).

Consequently, we have

$$2.6. \quad h_a + h_b + h_c = \frac{r^2 + s^2 + 4Rr}{2R},$$

$$2.7. \quad h_a h_b + h_b h_c + h_c h_a = \frac{2rs^2}{R},$$

$$2.8. \quad h_a h_b h_c = \frac{2r^2 s^2}{R}.$$

2.4 Cubic Equation for Reciprocals of Altitudes

We use (3) to obtain the cubic equation for the reciprocals of altitudes, namely $\frac{1}{h_a}$, $\frac{1}{h_b}$ and $\frac{1}{h_c}$ are the roots of the cubic equation in x given by

$$4r^2 s^2 x^3 - 4rs^2 x^2 + (r^2 + s^2 + 4Rr)x - 2R = 0. \quad (4)$$

Consequently, we have

$$2.9. \quad \frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} = \frac{1}{r},$$

$$2.10. \quad \frac{1}{h_a h_b} + \frac{1}{h_b h_c} + \frac{1}{h_c h_a} = \frac{r^2 + s^2 + 4Rr}{4r^2 s^2}.$$

2.5 Cubic Equation for $s - a, s - b, s - c$

We rewrite (1) as

$$(s - x)^3 - s(s - x)^2 + (r^2 + 4Rr)(s - x) - r^2 s = 0.$$

Since a, b, c are the roots of this equation in x , we see that $s - a, s - b, s - c$ are the roots of the equation in x given by

$$x^3 - sx^2 + (r^2 + 4Rr)x - r^2 s = 0. \quad (5)$$

Consequently, we have

$$2.11. \quad s - a + s - b + s - c = s,$$

$$2.12. \quad (s - a)(s - b) + (s - b)(s - c) + (s - c)(s - a) = r^2 + 4Rr,$$

$$2.13. \quad (s - a)(s - b)(s - c) = r^2 s.$$

2.6 Cubic Equation for Reciprocals of $s - a, s - b, s - c$

We use (5) to obtain the cubic equation for the reciprocals of $s - a, s - b, s - c$, namely $\frac{1}{s - a}, \frac{1}{s - b}$ and $\frac{1}{s - c}$ are the roots of the cubic equation in x given by

$$r^2 s x^3 - (r^2 + 4Rr)x^2 + sx - 1 = 0. \quad (6)$$

Consequently, we have

$$\mathbf{2.14.} \quad \frac{1}{s - a} + \frac{1}{s - b} + \frac{1}{s - c} = \frac{4R + r}{rs},$$

$$\mathbf{2.15.} \quad \frac{1}{(s - a)(s - b)} + \frac{1}{(s - b)(s - c)} + \frac{1}{(s - c)(s - a)} = \frac{1}{r^2}.$$

2.7 Cubic Equation for Exradii

We note that from (6) we have

$$r^2 s \left(\frac{1}{s - a} \right)^3 - (r^2 + 4Rr) \left(\frac{1}{s - a} \right)^2 + s \cdot \left(\frac{1}{s - a} \right) - 1 = 0.$$

We rewrite this as

$$r^2 \left(\frac{s}{s - a} \right)^3 - (r^2 + 4Rr) \left(\frac{s}{s - a} \right)^2 + s^2 \cdot \left(\frac{s}{s - a} \right) - s^2 = 0.$$

Then we use the fact that $\frac{s}{s - a} = \frac{r_a}{r}$ to obtain

$$r_a^3 - (4R + r)r_a^2 + s^2 r_a - rs^2 = 0$$

and we see that r_a is a root of the cubic equation in x given by

$$x^3 - (4R + r)x^2 + s^2 x - rs^2 = 0. \quad (7)$$

Similarly, r_b and r_c are roots of (7). Consequently, we have

$$\mathbf{2.16.} \quad r_a + r_b + r_c = 4R + r,$$

$$\mathbf{2.17.} \quad r_a r_b + r_b r_c + r_c r_a = s^2,$$

$$\mathbf{2.18.} \quad r_a r_b r_c = rs^2.$$

2.8 Cubic Equation for Reciprocals of Exradii

We use (7) to obtain the cubic equation for the reciprocals of exradii, namely $\frac{1}{r_a}, \frac{1}{r_b}$ and $\frac{1}{r_c}$ are the roots of the cubic equation in x given by

$$rs^2 x^3 - s^2 x^2 + (4R + r)x - 1 = 0. \quad (8)$$

Consequently, we have

$$\mathbf{2.19.} \quad \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c} = \frac{1}{r},$$

$$\mathbf{2.20.} \quad \frac{1}{r_a r_b} + \frac{1}{r_b r_c} + \frac{1}{r_c r_a} = \frac{4R + r}{rs^2}.$$

3 Some Geometric Inequalities

3.1 Trigonometric Inequalities

3.1. $\sin \frac{A}{2} \leq \frac{a}{2\sqrt{bc}}.$

PROOF. By the Cosine Rule,

$$a^2 = b^2 + c^2 - 2bc \cos A = (b - c)^2 + 4bc \sin^2 \frac{A}{2} \geq 4bc \sin^2 \frac{A}{2}.$$

The result follows by taking square roots and by rewriting. □

3.2. $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}.$

PROOF. We use the previous inequality to obtain

$$\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{a}{2\sqrt{bc}} \cdot \frac{b}{2\sqrt{ca}} \cdot \frac{c}{2\sqrt{ab}} = \frac{abc}{8\sqrt{a^2b^2c^2}} = \frac{1}{8}.$$

□

3.3. $\cos A + \cos B + \cos C \leq \frac{3}{2}.$

PROOF. We use the previous inequality to obtain

$$\begin{aligned} \cos A + \cos B + \cos C &= 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) + 1 - 2 \sin^2 \frac{C}{2} \\ &= 2 \sin \frac{C}{2} \cos \left(\frac{A-B}{2} \right) + 1 - 2 \sin^2 \frac{C}{2} \\ &= 1 + 2 \sin \frac{C}{2} \left(\cos \left(\frac{A-B}{2} \right) - \sin \frac{C}{2} \right) \\ &= 1 + 2 \sin \frac{C}{2} \left(\cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right) \\ &= 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\ &\leq 1 + 4 \cdot \frac{1}{8} \\ &= \frac{3}{2}. \end{aligned}$$

□

3.4. $\cos A \cos B \cos C \leq \frac{1}{8}.$

PROOF. For non acute-angled triangles, the inequality is obvious. For acute angled-triangles, we use the AM-GM Inequality and the previous inequality to obtain

$$\cos A \cos B \cos C \leq \left(\frac{\cos A + \cos B + \cos C}{3} \right)^3 \leq \left(\frac{1}{3} \cdot \frac{3}{2} \right)^3 = \frac{1}{8}.$$

□

3.5. $\sin \frac{A}{2} + \sin \frac{B}{2} + \sin \frac{C}{2} \leq \frac{3}{2}.$

PROOF. If $A' = 90^\circ - \frac{A}{2}$, $B' = 90^\circ - \frac{B}{2}$, $C' = 90^\circ - \frac{C}{2}$, then $A' + B' + C' = 180^\circ$ and so there is a triangle with angles A', B', C' . Therefore

$$\sin \frac{A}{2} + \sin \frac{B}{2} + \sin \frac{C}{2} = \cos A' + \cos B' + \cos C' \leq \frac{3}{2}.$$

□

3.6. $\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \leq \frac{3\sqrt{3}}{8}.$

PROOF. Observe that

$$\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} = \frac{1}{2} (1 + \cos A + 1 + \cos B + 1 + \cos C) \leq \frac{1}{2} \left(3 + \frac{3}{2} \right) = \frac{9}{4}.$$

Therefore by the AM-GM Inequality,

$$\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \leq \left(\frac{1}{3} \left(\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} \right) \right)^{\frac{3}{2}} \leq \left(\frac{1}{3} \left(\frac{9}{4} \right) \right)^{\frac{3}{2}} = \frac{3\sqrt{3}}{8}$$

□

3.2 Inequalities Involving Elements of a Triangle

3.7 (Euler's Inequality). $R \geq 2r$.

PROOF. Observe that $R(R - 2r) = OI^2 \geq 0$. □

3.8. $a + b + c \leq 3\sqrt{3}R$.

PROOF. By the Cauchy-Schwarz Inequality, we have

$$a + b + c \leq \sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{a^2 + b^2 + c^2} \leq 3\sqrt{3}R.$$

because $9R^2 - (a^2 + b^2 + c^2) = OH^2 \geq 0$.

For completeness, we show that $OH^2 = 9R^2 - (a^2 + b^2 + c^2)$.

If $\triangle ABC$ is non-obtuse-angled, we have by the Cosine Formula and the Sine Formula,

$$\begin{aligned} OH^2 &= AO^2 + AH^2 - 2 \cdot AO \cdot AH \cdot \cos |B - C| \\ &= R^2 + 4R^2 \cos^2 A - 4R^2 \cos A \cos(B - C) \\ &= 5R^2 - 4R^2 \sin^2 A + 4R^2 \cos(B + C) \cos(B - C) \\ &= 5R^2 - 4R^2 \sin^2 A + 4R^2 (1 - \sin^2 B - \sin^2 C) \\ &= 9R^2 - 4R^2 (\sin^2 A + \sin^2 B + \sin^2 C) \\ &= 9R^2 - (a^2 + b^2 + c^2). \end{aligned}$$

If $\triangle ABC$ is obtuse-angled, say at A , we have $AH = -2R \cos A$, and again by the Cosine formula and the Sine Formula,

$$OH^2 = AO^2 + AH^2 - 2 \cdot AO \cdot AH \cdot \cos(180^\circ + |B - C|) = 9R^2 - (a^2 + b^2 + c^2).$$

□

3.9 (Weitzenböck's Inequality). $a^2 + b^2 + c^2 \geq 4\sqrt{3}\Delta$.

PROOF. By Heron's Formula, we have

$$16\Delta^2 = (a + b + c)(a + b - c)(b + c - a)(c + a - b) \leq (a + b + c) \left(\frac{a + b + c}{3} \right)^3 = \frac{(a + b + c)^4}{27},$$

and so, by the Cauchy-Schwarz Inequality,

$$4\sqrt{3}\Delta \leq \frac{(a + b + c)^2}{3} \leq \frac{(1^2 + 1^2 + 1^2)(a^2 + b^2 + c^2)}{3} = a^2 + b^2 + c^2.$$

□

3.10 (Hadwiger-Finsler Inequality). $a^2 + b^2 + c^2 \geq 4\sqrt{3}\Delta + (a - b)^2 + (b - c)^2 + (c - a)^2$.

PROOF. Observe that

$$\begin{aligned} a^2 + b^2 + c^2 - ((a - b)^2 + (b - c)^2 + (c - a)^2) &= (a^2 - (b - c)^2) + (b^2 - (c - a)^2) + (c^2 - (a - b)^2) \\ &= 4((s - b)(s - c) + (s - c)(s - a) + (s - a)(s - b)) \end{aligned}$$

and, by Heron's Formula,

$$4\sqrt{3}\Delta = 4\sqrt{3s(s - a)(s - b)(s - c)}.$$

Hence, if we let $x = s - a$, $y = s - b$, $z = s - c$, the desired inequality can be rewritten as

$$xy + yz + zx \geq \sqrt{3(x + y + z)xyz}$$

or equivalently,

$$x^2y^2 + y^2z^2 + z^2x^2 \geq x^2yz + xy^2z + xyz^2,$$

which is obviously true because

$$(xy - zx)^2 + (yz - xy)^2 + (zx - yz)^2 \geq 0.$$

□

3.11. $ab + bc + ca \geq 4\sqrt{3}\Delta$.

PROOF. By the AM-GM Inequality, we have

$$ab + bc + ca = 2\Delta \left(\frac{1}{\sin C} + \frac{1}{\sin A} + \frac{1}{\sin B} \right) \geq 6\Delta \left(\frac{1}{\sin C} \cdot \frac{1}{\sin A} \cdot \frac{1}{\sin B} \right)^{\frac{1}{3}} \geq 6\Delta \left(\frac{8}{3\sqrt{3}} \right)^{\frac{1}{3}} = 4\sqrt{3}\Delta$$

because, again by the AM-GM Inequality,

$$6R\sqrt[3]{\sin A \sin B \sin C} \leq 2R(\sin A + \sin B + \sin C) = a + b + c \leq 3\sqrt{3}R$$

and so

$$\frac{1}{\sin A \sin B \sin C} \geq \left(\frac{2}{\sqrt{3}} \right)^3 = \frac{8}{3\sqrt{3}}.$$

□

3.12. $a^4 + b^4 + c^4 \geq 16\Delta^2$.

PROOF. By the Cauchy-Schwarz Inequality and Weitzenböck's Inequality, we have

$$a^4 + b^4 + c^4 \geq \frac{1}{3} (a^2 + b^2 + c^2)^2 \geq \frac{1}{3} (4\sqrt{3}\Delta)^2 = 16\Delta^2.$$

□

3.13. $\frac{a^2}{r_b r_c} + \frac{b^2}{r_c r_a} + \frac{c^2}{r_a r_b} \geq 4$.

PROOF. By the Cauchy-Schwarz Inequality, we have

$$\frac{a^2}{r_b r_c} + \frac{b^2}{r_c r_a} + \frac{c^2}{r_a r_b} \geq \frac{(a + b + c)^2}{r_b r_c + r_c r_a + r_a r_b} = \frac{4s^2}{s^2} = 4.$$

□

3.14. $\frac{r_a^2}{a^2 + r_a^2} + \frac{r_b^2}{b^2 + r_b^2} + \frac{r_c^2}{c^2 + r_c^2} \geq \frac{4R + r}{4R - r}$.

PROOF. By the Cauchy-Schwarz Inequality, we have

$$\begin{aligned} \frac{r_a^2}{a^2 + r_a^2} + \frac{r_b^2}{b^2 + r_b^2} + \frac{r_c^2}{c^2 + r_c^2} &\geq \frac{(r_a + r_b + r_c)^2}{a^2 + r_a^2 + b^2 + r_b^2 + c^2 + r_c^2} \\ &= \frac{(r_a + r_b + r_c)^2}{a^2 + b^2 + c^2 + r_a^2 + r_b^2 + r_c^2} \\ &= \frac{(4R + r)^2}{(2s)^2 - 2(r^2 + s^2 + 4Rr) + (4R + r)^2 - 2s^2} \\ &= \frac{(4R + r)^2}{(4R - r)(4R + r)} \\ &= \frac{4R + r}{4R - r}. \end{aligned}$$

□

3.15. $5R - r \geq \sqrt{3}s$.

PROOF. By rewriting the Hadwiger-Finsler Inequality, we have

$$2(bc + ca + ab) - (a^2 + b^2 + c^2) \geq 4\sqrt{3}\Delta.$$

Hence, by Euler's Inequality and the fact that

$$r_a(s - a) = r_b(s - b) = r_c(s - c) = rs,$$

we have

$$\begin{aligned} 5R - r &= 4R + r + (R - 2r) \\ &\geq 4R + r \\ &= r_a + r_b + r_c \\ &= rs \left(\frac{1}{s - a} + \frac{1}{s - b} + \frac{1}{s - c} \right) \\ &= \frac{s}{4\Delta} (2(bc + ca + ab) - (a^2 + b^2 + c^2)) \\ &\geq \frac{s}{4\Delta} \cdot 4\sqrt{3}\Delta \\ &= \sqrt{3}s. \end{aligned}$$

□

3.16. $(R + r)^2 \geq \sqrt{3}\Delta$.

PROOF. Using the fact that $a + b + c \leq 3\sqrt{3}R$ and the AM-GM Inequality, we have

$$R + r \geq \frac{1}{3} \left(\frac{s}{\sqrt{3}} + \frac{s}{\sqrt{3}} + 3r \right) \geq \sqrt[3]{rs^2} = \sqrt[3]{s\Delta},$$

and so

$$(R + r)^2 \geq \sqrt[3]{s^2\Delta^2} \geq \sqrt[3]{3\sqrt{3}\Delta \cdot \Delta^2} = \sqrt{3}\Delta,$$

because, by the AM-GM Inequality,

$$\frac{\Delta^2}{s} = (s - a)(s - b)(s - c) \leq \left(\frac{s - a + s - b + s - c}{3} \right)^3 = \frac{s^3}{27},$$

that is, $s^2 \geq 3\sqrt{3}\Delta$. □

3.17. $(a + b + c)(h_a + h_b + h_c) \geq 18\Delta$.

PROOF. If $a \leq b \leq c$, then $h_c \leq h_b \leq h_a$, and so by Chebyshev's Inequality,

$$(a + b + c)(h_a + h_b + h_c) \geq 3(ah_a + bh_b + ch_c) = 3 \cdot (6\Delta) = 18\Delta. \quad \square$$

3.18. $m_a + m_b + m_c \leq 4R + r$.

PROOF. If $\triangle ABC$ is acute-angled, we have

$$m_a + m_b + m_c \leq R + p_a + R + p_b + R + p_c = 3R + R(\cos A + \cos B + \cos C) = 3R + R \left(1 + \frac{r}{R} \right) = 4R + r,$$

because

$$\begin{aligned} \cos A + \cos B + \cos C - 1 &= 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\ &= 4 \sqrt{\frac{(s-b)(s-c)}{bc}} \sqrt{\frac{(s-c)(s-a)}{ca}} \sqrt{\frac{(s-a)(s-b)}{ab}} \\ &= 4 \cdot \frac{(s-a)(s-b)(s-c)}{abc} \\ &= 4 \cdot \frac{\Delta^2}{s \cdot 4R\Delta} \\ &= \frac{r}{R}. \end{aligned}$$

In any $\triangle ABC$, we use the following facts:

- $ab + bc + ca = r^2 + s^2 + 4Rr$,
- $a^2 + b^2 + c^2 = (2s)^2 - 2(r^2 + s^2 + 4Rr) = 2s^2 - 2r^2 - 8Rr$,

and so, by Gerretsen's Inequality and Euler's Inequality, we have

$$\begin{aligned} (m_a + m_b + m_c)^2 &= m_a^2 + m_b^2 + m_c^2 + 2(m_am_b + m_bm_c + m_cm_a) \\ &\leq \frac{3}{4} (a^2 + b^2 + c^2) + \left(c^2 + \frac{ab}{2} + a^2 + \frac{bc}{2} + b^2 + \frac{ca}{2} \right) \\ &= 4s^2 - 3r^2 - 12Rr \\ &\leq 4(4R^2 + 4Rr + 3r^2) - 3r^2 - 12Rr \\ &= 16R^2 + 4Rr + 9r^2 \\ &= (4R + r)^2 - 4r(R - 2r) \\ &\leq (4R + r)^2, \end{aligned}$$

because

$$\begin{aligned} 2m_am_b &\leq c^2 + \frac{ab}{2} \\ \iff (2b^2 + 2c^2 - a^2)(2c^2 + 2a^2 - b^2) &\leq (2c^2 + ab)^2 \\ \iff 2a^4 + 2b^4 - 4a^2b^2 - 2b^2c^2 - 2c^2a^2 - 4abc^2 &\geq 0 \\ \iff 2(a^2 - b^2)^2 - 2(ca - bc)^2 &\geq 0 \\ \iff 2(a - b)^2(a + b + c)(a + b - c) &\geq 0, \end{aligned}$$

which is true by the Triangle Inequality. Similarly, $2m_bm_c \leq a^2 + \frac{bc}{2}$ and $2m_cm_a \leq b^2 + \frac{ca}{2}$. □

3.19. $m_a^2 + m_b^2 + m_c^2 \geq 3\sqrt{3}\Delta.$

PROOF. By Weitzenböck's Inequality, we have

$$\begin{aligned} m_a^2 + m_b^2 + m_c^2 &= \frac{1}{4}(2b^2 + 2c^2 - a^2) + \frac{1}{4}(2c^2 + 2a^2 - b^2) + \frac{1}{4}(2a^2 + 2b^2 - c^2) \\ &= \frac{3}{4}(a^2 + b^2 + c^2) \\ &\geq \frac{3}{4} \cdot 4\sqrt{3}\Delta \\ &= 3\sqrt{3}\Delta, \end{aligned}$$

because, by the Cosine Formula, we have

$$m_a^2 = \frac{a^2}{4} + b^2 - 2 \cdot \frac{a}{2} \cdot b \cdot \left(\frac{a^2 + b^2 - c^2}{2ab} \right) = \frac{1}{4}(2b^2 + 2c^2 - a^2),$$

and, similarly,

$$m_b^2 = \frac{1}{4}(2c^2 + 2a^2 - b^2) \quad \text{and} \quad m_c^2 = \frac{1}{4}(2a^2 + 2b^2 - c^2).$$

□

3.20. $\frac{a}{b+c+2m_a} + \frac{b}{c+a+2m_b} + \frac{c}{a+b+2m_c} \geq \frac{b+c-2m_a}{a} + \frac{c+a-2m_b}{b} + \frac{a+b-2m_c}{c}.$

PROOF. Observe that

$$a^2 \geq a^2 - (b-c)^2 = (b^2 + 2bc + c^2) - (2b^2 + 2c^2 - a^2) = (b+c)^2 - 4m_a^2,$$

and so

$$\frac{a}{b+c+2m_a} \geq \frac{b+c-2m_a}{a}.$$

Similarly,

$$\frac{b}{c+a+2m_b} \geq \frac{c+a-2m_b}{b} \quad \text{and} \quad \frac{c}{a+b+2m_c} \geq \frac{a+b-2m_c}{c}.$$

We add them to obtain the desired inequality.

□

3.21. $\frac{1}{m_a} + \frac{1}{m_b} + \frac{1}{m_c} \leq \frac{R}{2r^2}.$

PROOF. Since $m_a \geq h_a$, $m_b \geq h_b$, $m_c \geq h_c$, it follows that, by Euler's Inequality,

$$\frac{1}{m_a} + \frac{1}{m_b} + \frac{1}{m_c} \leq \frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} = \frac{1}{r} \leq \frac{R}{2r^2}$$

□

3.22. If $\triangle ABC$ is acute-angled, then $\frac{m_a}{r_a} + \frac{m_b}{r_b} + \frac{m_c}{r_c} \leq \frac{2R}{r} - 1.$

PROOF. Since

$$m_a \leq R + p_a = R(1 + \cos A), \quad m_b \leq R + p_b = R(1 + \cos B), \quad m_c \leq R + p_c = R(1 + \cos C)$$

and

$$rs = r_a(s-a) = r_b(s-b) = r_c(s-c),$$

we have

$$\begin{aligned} \frac{m_a}{r_a} + \frac{m_b}{r_b} + \frac{m_c}{r_c} &\leq \frac{R}{rs} \left((1 + \cos A)(s-a) + (1 + \cos B)(s-b) + (1 + \cos C)(s-c) \right) \\ &= \frac{R}{r} (3 + \cos A + \cos B + \cos C) - \frac{R}{rs} (2s + a \cos A + b \cos B + c \cos C) \\ &= \frac{R}{r} (3 + \cos A + \cos B + \cos C) - \frac{2R}{r} - \frac{R}{rs} (a \cos A + b \cos B + c \cos C) \\ &= \frac{R}{r} + \frac{R}{r} \left(1 + \frac{r}{R} \right) - \frac{R}{rs} \left(\frac{2\Delta}{R} \right) \\ &= \frac{2R}{r} + 1 - 2 \\ &= \frac{2R}{r} - 1. \end{aligned}$$

□

3.23. $r_a r_b r_c \leq \frac{3abc\sqrt{3}}{8}.$

PROOF. Using the fact that $a+b+c \leq 3\sqrt{3}R$, we have

$$r_a r_b r_c = \frac{\Delta^2}{(s-a)(s-b)(s-c)} = s\Delta = \frac{a+b+c}{2} \cdot \frac{abc}{4R} \leq 3\sqrt{3}R \cdot \frac{abc}{8R} = \frac{3abc\sqrt{3}}{8}.$$

□

3.24 (Carlitz's Inequality). $w_a w_b w_c \leq r s^2$.

PROOF. By the AM-GM Inequality, we have

$$\begin{aligned} w_a w_b w_c &= \frac{2\sqrt{bc}}{b+c} \sqrt{s(s-a)} \cdot \frac{2\sqrt{ca}}{c+a} \sqrt{s(s-b)} \cdot \frac{2\sqrt{ab}}{a+b} \sqrt{s(s-c)} \\ &\leq \frac{2\sqrt{bc}}{2\sqrt{bc}} \sqrt{s(s-a)} \cdot \frac{2\sqrt{ca}}{2\sqrt{ca}} \sqrt{s(s-b)} \cdot \frac{2\sqrt{ab}}{2\sqrt{ab}} \sqrt{s(s-c)} \\ &= \sqrt{s^3(s-a)(s-b)(s-c)} \\ &= r s^2, \end{aligned}$$

because by considering areas, we have

$$bc \sin \frac{A}{2} \cos \frac{A}{2} = \frac{1}{2} bc \sin A = \frac{1}{2} b w_a \sin \frac{A}{2} + \frac{1}{2} c w_a \sin \frac{A}{2}$$

and so

$$w_a = \frac{2bc}{b+c} \cos \frac{A}{2} = \frac{2\sqrt{bc}}{b+c} \sqrt{s(s-a)},$$

and similarly,

$$w_b = \frac{2\sqrt{ca}}{c+a} \sqrt{s(s-b)}, \quad w_c = \frac{2\sqrt{ab}}{a+b} \sqrt{s(s-c)}.$$

□

3.25. $w_a + w_b + w_c \leq \sqrt{3}s$.

PROOF. Observe that, by the AM-GM Inequality, we have

$$w_a = \frac{2\sqrt{bc}}{b+c} \sqrt{s(s-a)} \leq \frac{2\sqrt{bc}}{2\sqrt{bc}} \sqrt{s(s-a)} = \sqrt{s(s-a)}.$$

Similarly,

$$w_b \leq \sqrt{s(s-b)} \quad \text{and} \quad w_c \leq \sqrt{s(s-c)}.$$

Therefore, by the Cauchy-Schwarz Inequality, we have

$$w_a + w_b + w_c \leq \sqrt{s(s-a)} + \sqrt{s(s-b)} + \sqrt{s(s-c)} \leq \sqrt{s+s+s} \sqrt{s-a+s-b+s-c} = \sqrt{3}s.$$

□

3.26. $h_a + h_b + h_c \leq \sqrt{3}s$.

PROOF. Since $w_a \geq h_a$, $w_b \geq h_b$, $w_c \geq h_c$, we have

$$h_a + h_b + h_c \leq w_a + w_b + w_c \leq \sqrt{3}s.$$

□

3.27. $w_a^2 + w_b^2 + w_c^2 \geq 3\sqrt{3}\Delta$.

PROOF. By Stewart's Theorem,

$$b^2 \cdot \frac{ca}{b+c} + c^2 \cdot \frac{ba}{b+c} = a \left(w_a^2 + \frac{ca}{b+c} \cdot \frac{ba}{b+c} \right),$$

and so

$$w_a^2 = bc - \frac{a^2 bc}{(b+c)^2} \geq bc - \frac{a^2}{4}$$

because $\frac{bc}{(b+c)^2} \leq \frac{1}{4}$. Similarly,

$$w_b^2 \geq ca - \frac{b^2}{4} \quad \text{and} \quad w_c^2 \geq ab - \frac{c^2}{4}.$$

Now, by rewriting the Hadwiger-Finsler Inequality, we have

$$\frac{1}{4} (a^2 + b^2 + c^2) \leq \frac{1}{2} (ab + bc + ca) - \sqrt{3}\Delta,$$

and so

$$\begin{aligned} w_a^2 + w_b^2 + w_c^2 &\geq ab + bc + ca - \frac{1}{4} (a^2 + b^2 + c^2) \\ &\geq ab + bc + ca - \left(\frac{1}{2} (ab + bc + ca) - \sqrt{3}\Delta \right) \\ &= \frac{1}{2} (ab + bc + ca) + \sqrt{3}\Delta \\ &\geq \frac{1}{2} \cdot 4\sqrt{3}\Delta + \sqrt{3}\Delta \\ &= 3\sqrt{3}\Delta. \end{aligned}$$

□

3.28. If $\triangle ABC$ is acute-angled, then $\frac{m_a m_b m_c}{r_a r_b r_c} + \frac{w_a w_b w_c}{h_a h_b h_c} \leq \frac{R}{r}$.

PROOF. Observe that

$$m_a \leq R + p_a = R(1 + \cos A) = 2R \cos^2 \frac{A}{2},$$

and similarly,

$$m_b \leq 2R \cos^2 \frac{B}{2} \quad m_c \leq 2R \cos^2 \frac{C}{2}.$$

Since

$$m_a m_b m_c \leq 8 \cos^2 \frac{A}{2} \cos^2 \frac{B}{2} \cos^2 \frac{C}{2} = 8R^3 \cdot \frac{s(s-a)}{bc} \cdot \frac{s(s-b)}{ca} \cdot \frac{s(s-c)}{ab} = 8R^3 \cdot \frac{s^2 \Delta^2}{16R^2 \Delta^2} = \frac{rs^2 R}{2r},$$

and

$$r_a r_b r_c = \frac{\Delta^2}{(s-a)(s-b)(s-c)} = \frac{s\Delta^3}{\Delta^2} = rs^2,$$

we have

$$\frac{m_a m_b m_c}{r_a r_b r_c} \leq \frac{rs^2 R}{2r} \cdot \frac{1}{rs^2} = \frac{R}{2r}.$$

Now, by Carlitz's Inequality,

$$\frac{w_a w_b w_c}{h_a h_b h_c} \leq \frac{rs^2}{\frac{2\Delta}{a} \cdot \frac{2\Delta}{b} \cdot \frac{2\Delta}{c}} = \frac{rs^2 \cdot 4R\Delta}{8\Delta^3} = \frac{rs^2 R\Delta}{2\Delta^3} = \frac{sR\Delta^2}{2\Delta^3} = \frac{R}{2r}.$$

Therefore

$$\frac{m_a m_b m_c}{r_a r_b r_c} + \frac{w_a w_b w_c}{h_a h_b h_c} \leq \frac{R}{2r} + \frac{R}{2r} = \frac{R}{r}.$$

□

3.29. $36Rr + 9r^2 \leq 3s^2 \leq (4R + r)^2$.

PROOF. Recall that from (1) and (7)

- a, b, c are the roots of cubic equation in x given by

$$x^3 - 2sx^2 + (r^2 + s^2 + 4Rr)x - 4Rrs = 0;$$

- r_a, r_b, r_c are the roots of cubic equation in x given by

$$x^3 - (4R + r)x^2 + s^2x - rs^2 = 0.$$

By taking the derivative, both equations

$$3x^2 - 4sx + (r^2 + s^2 + 4Rr) = 0 \quad \text{and} \quad 3x^2 - 2(4R + r)x + s^2 = 0$$

have real roots because of Rolle's Theorem. Hence by considering their discriminants, we get

$$4(s^2 - (12Rr + 3r^2)) = (-4s)^2 - 4 \cdot 3 \cdot (r^2 + s^2 + 4Rr) \geq 0$$

and

$$4((4R + r)^2 - 3s^2) = (2(4R + r))^2 - 4 \cdot 3s^2 \geq 0,$$

which prove the inequalities.

□

3.30 (Gerretsen's Inequality). $16Rr - 5r^2 \leq s^2 \leq 4R^2 + 4Rr + 3r^2$.

PROOF 1. The right side inequality follows from the fact that

$$4R^2 + 4Rr + 3r^2 - s^2 = HI^2 \geq 0, \quad (9)$$

and the left side inequality follows from the fact that

$$s^2 - (16Rr - 5r^2) = 9 \cdot IG^2 \geq 0. \quad (10)$$

Now we prove the formulas in (9) and (10) by using the Cosine Formula.

If $\triangle ABC$ is acute-angled, we use the following facts:

- $\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \sqrt{\frac{s(s-a)}{bc}} \cdot \sqrt{\frac{s(s-b)}{ca}} \cdot \sqrt{\frac{s(s-c)}{ca}} = \frac{s\Delta}{abc} = \frac{s}{4R},$
- $\cos A \cos B \cos C = \frac{R^2 - OH^2}{8R^2} = \frac{s^2 - (2R+r)^2}{4R^2},$
- $AI = \frac{r}{\sin \frac{A}{2}} = \frac{\frac{abc}{4Rs}}{\sin \frac{A}{2}} = \frac{2R^2 \sin A \sin B \sin C}{s \sin \frac{A}{2}} = \frac{16R^2 \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{B}{2} \sin \frac{C}{2} \cos \frac{C}{2}}{s} = 4R \sin \frac{B}{2} \sin \frac{C}{2},$

and so we have

$$\begin{aligned} HI^2 &= 4R^2 \cos^2 A + 16R^2 \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} - 16R^2 \cos A \sin \frac{B}{2} \sin \frac{C}{2} \cos \left(\frac{B}{2} - \frac{C}{2} \right) \\ &= 4R^2 \left(\cos^2 A + 4 \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} - \cos A \cdot 2 \sin \frac{B}{2} \cos \frac{B}{2} \cdot 2 \sin \frac{C}{2} \cos \frac{C}{2} - 4 \cos A \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} \right) \\ &= 4R^2 \left(\cos^2 A + 8 \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} \cdot \frac{1}{2} (1 - \cos A) - \cos A \sin B \sin C \right) \\ &= 4R^2 \left(8 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} + \cos A (\cos(180^\circ - (B+C)) - \sin B \sin C) \right) \\ &= 4R^2 \left(8 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} - \cos A \cos B \cos C \right) \\ &= 2r^2 - 4R^2 \cos A \cos B \cos C \\ &= 2r^2 - 4R^2 \left(\frac{s^2 - (2R+r)^2}{4R^2} \right) \\ &= 4R^2 + 4Rr + 3r^2 - s^2. \end{aligned}$$

If $\triangle ABC$ is non-acute-angled, say at A , we have $AH = -2R \cos A$, and we have

$$HI^2 = 4R^2 \cos^2 A + 16R^2 \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} + 16R^2 \cos A \sin \frac{B}{2} \sin \frac{C}{2} \cos \left(180^\circ + \frac{B}{2} - \frac{C}{2} \right)$$

which yields the same result.

To prove the left side inequality, we use the facts that H, G, O are collinear and $OH = 3 \cdot OG$ to obtain

$$\begin{aligned} 9 \cdot IG^2 &= 9(OI^2 + OG^2 - 2 \cdot OI \cdot OG \cos \angle HOI) \\ &= 9 \cdot OI^2 + OH^2 - 6 \cdot OI \cdot OH \left(\frac{OI^2 + OH^2 - HI^2}{2 \cdot OI \cdot OH} \right) \\ &= 9 \cdot OI^2 + OH^2 - 3(OI^2 + OH^2 - HI^2) \\ &= 6 \cdot OI^2 + 3 \cdot HI^2 - 2 \cdot OH^2 \\ &= 6(R^2 - 2Rr) + 3(4R^2 + 4Rr + 3r^2) - 2(9R^2 - (2s^2 - 2r^2 - 8Rr)) \\ &= s^2 - (16Rr - 5r^2). \end{aligned}$$

□

PROOF 2. Let $x = s - a, y = s - b, z = s - c$.

We rewrite the left-side inequality as $s^2 - 16Rr + 5r^2 \geq 0$ which is equivalent to

$$x(x-y)(x-z) + y(y-z)(y-x) + z(z-x)(z-y) \geq 0$$

and this is true by Schur's Inequality.

We rewrite the right-side inequality as $4R^2 + 4Rr + 3r^2 - s^2 \geq 0$ which (after some really tedious manipulations) is equivalent to

$$x^4(y-z)^2 + y^4(z-x)^2 + z^4(x-y)^2 + 2(xy(xy-yz)(xy-zx) + yz(yz-zx)(yz-xy) + zx(zx-xy)(zx-yz)) \geq 0$$

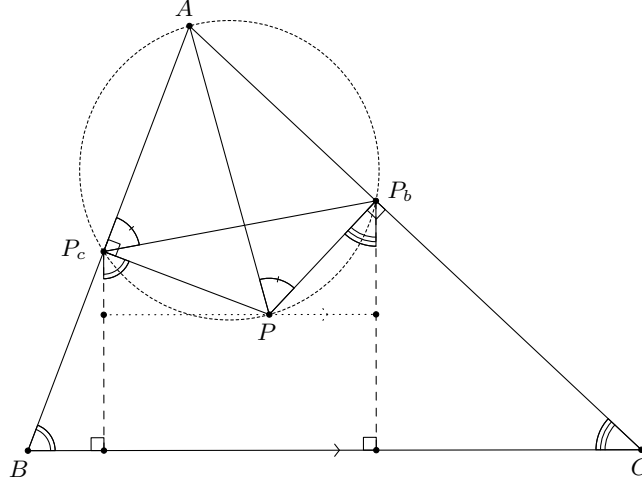
and again this is true by Schur's Inequality.

□

3.3 Erdős-Mordell Inequality and Some Related Inequalities

3.31 (Mordell's Lemma). $R_1 \geq \frac{c}{a}r_2 + \frac{b}{a}r_3$.

PROOF 1.



By the Sine Formula, we rewrite the desired inequality as

$$R_1 \sin A \geq r_2 \sin C + r_3 \sin B. \quad (*)$$

By the Sine Formula (on $\triangle AP_cP_b$) and the fact that $\angle AP_cP_b = \angle APP_b$, we have

$$\frac{P_bP_c}{\sin A} = \frac{AP_b}{\sin \angle AP_cP_b} = \frac{AP_b}{\sin \angle APP_b} = R_1,$$

where the last equality is due to the fact that $\triangle APP_b$ is right-angled.

Therefore

$$R_1 \sin A = P_bP_c \geq r_2 \sin C + r_3 \sin B,$$

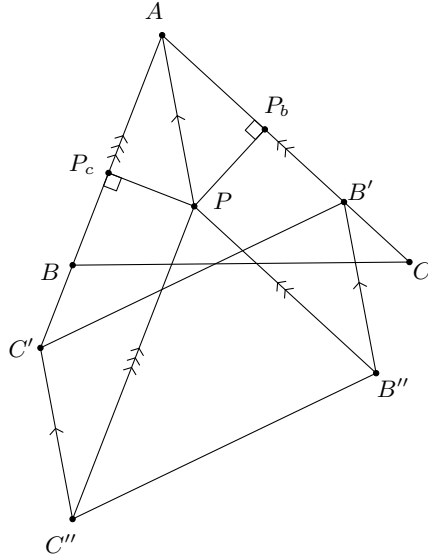
because the right-hand side is nothing but the length of the orthogonal projection of P_bP_c onto BC (see figure above for the case of an acute-angled triangle). $\square$

PROOF 2. As in the previous proof, we have $P_bP_c = R_1 \sin A$. Now, by the Cosine Formula (applied to $\triangle PP_bP_c$), we have

$$\begin{aligned} P_bP_c^2 &= r_2^2 + r_3^2 - 2r_2r_3 \cos(180^\circ - A) \\ &= r_2^2 + r_3^2 - 2r_2r_3 \cos(B + C) \\ &= r_2^2 + r_3^2 - 2r_2r_3 (\cos B \cos C - \sin B \sin C) \\ &= (r_2 \sin C + r_3 \sin B)^2 + (r_2 \cos C - r_3 \cos B)^2 \\ &\geq (r_2 \sin C + r_3 \sin B)^2. \end{aligned}$$

The result follows. $\square$

PROOF 3.

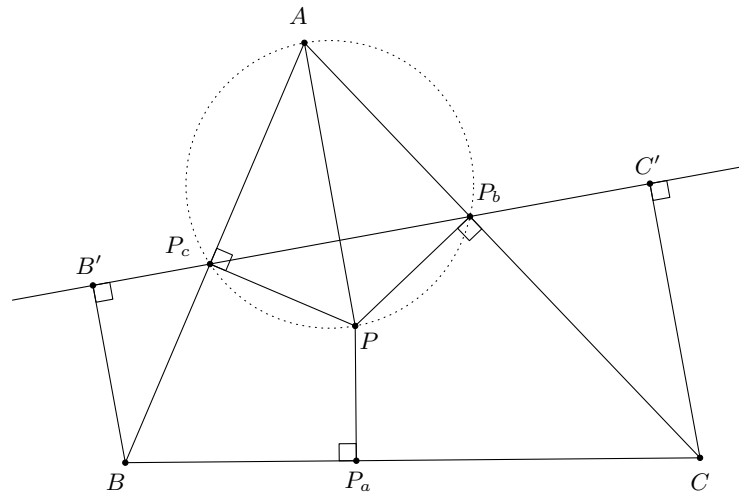


We choose points C' and B' on the sidelines AB and AC , respectively, such that $AC' = AC$ and $AB' = AB$. Let C'' and B'' be points on the same side as P with respect to AB and AC such that $AC'C''P$ and $APB''B'$ are parallelograms, respectively. Then $C'C''B''B'$ is a parallelogram. Since $\triangle ABC \cong \triangle AB'C'$, we have $BC = B'C'$. Hence

$$\begin{aligned}
 cr_2 + br_3 &= AC \cdot PP_c + AB \cdot PP_b \\
 &= AC' \cdot PP_c + AB' \cdot PP_b \\
 &= [APB''B'] + [AC'C''P] \\
 &= [C'C''B''B'] \\
 &\leq B'C' \cdot B'B'' \\
 &= BC \cdot PA \\
 &= a \cdot R_1.
 \end{aligned}$$

The result follows. □

PROOF 4.



Let $B'C'$ be the orthogonal projection of BC onto the line P_cP_b as shown above. Then

$$a = BC \geq B'C' = B'P_c + P_cP_b + P_bC'.$$

Since $\angle BP_cB' = \angle AP_cP_b = \angle APP_b$, then the right triangles BP_cB' and APP_b are similar and

$$B'P_c = \frac{PP_b}{PA} \cdot P_cB = \frac{r_2}{R_1} \cdot P_cB.$$

Similarly,

$$P_b C' = \frac{r_3}{R_1} \cdot C P_b.$$

By Ptolemy's Theorem (applied to $AP_c P P_b$),

$$R_1 \cdot P_c P_b = r_2 \cdot AP_c + r_3 \cdot AP_b,$$

or equivalently,

$$P_c P_b = \frac{r_2 \cdot AP_c + r_3 \cdot P_b A}{R_1}$$

We combine the results and get

$$a \geq \frac{r_2}{R_1} \cdot P_c B + \frac{r_2 \cdot AP_c + r_3 \cdot P_b A}{R_1} + \frac{r_3}{R_1} \cdot C P_b = \frac{r_2}{R_1} \cdot AB + \frac{r_3}{R_1} \cdot CA = \frac{r_2}{R_1} c + \frac{r_3}{R_1} b.$$

Rewrite and the result follows. $\square$

3.32 (Erdős-Mordell Inequality). $R_1 + R_2 + R_3 \geq 2(r_1 + r_2 + r_3)$.

PROOF. We use Mordell's Lemma and the AM-GM Inequality to obtain

$$\begin{aligned} R_1 + R_2 + R_3 &\geq \left(\frac{c}{a} r_2 + \frac{b}{a} r_3 \right) + \left(\frac{a}{b} r_3 + \frac{c}{b} r_1 \right) + \left(\frac{b}{c} r_1 + \frac{a}{c} r_2 \right) \\ &= \left(\frac{b}{c} + \frac{c}{b} \right) r_1 + \left(\frac{c}{a} + \frac{a}{c} \right) r_2 + \left(\frac{a}{b} + \frac{b}{a} \right) r_3 \\ &\geq 2r_1 \sqrt{\frac{b}{c} \cdot \frac{c}{b}} + 2r_2 \sqrt{\frac{c}{a} \cdot \frac{a}{c}} + 2r_3 \sqrt{\frac{a}{b} \cdot \frac{b}{a}} \\ &= 2(r_1 + r_2 + r_3). \end{aligned}$$

$\square$

3.33. If $\triangle ABC$ is acute-angled, $AO + BO + CO \geq AH + BH + CH$.

PROOF. Since $HA = 2p_a$, $HB = 2p_b$ and $HC = 2p_c$, we have, by the Erdős-Mordell Inequality,

$$AO + BO + CO \geq 2(p_a + p_b + p_c) = AH + BH + CH.$$

$\square$

Note that if $\triangle ABC$ is obtuse-angled, the inequality is false: consider an isosceles triangle with $\angle CAB = 120^\circ$, $\angle ABC = 30^\circ$ and $\angle BCA = 30^\circ$, and with circumradius R . Then $AH + BH + CH = R + R\sqrt{3} + R\sqrt{3} > 3R$.

3.34. If $\triangle ABC$ is acute-angled, then $h_a + h_b + h_c \leq 3(R + r)$.

PROOF. We use the following facts:

- $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$,
- $AH + BH + CH = 2(p_a + p_b + p_c) = 2R(\cos A + \cos B + \cos C) = 2R\left(1 + \frac{r}{R}\right) = 2(R + r)$.

Then, by the Erdős-Mordell Inequality, we have

$$\begin{aligned} h_a + h_b + h_c &= AH + BH + CH + (HH_a + HH_b + HH_c) \\ &\leq AH + BH + CH + \frac{1}{2}(AH + BH + CH) \\ &= \frac{3}{2}(AH + BH + CH) \\ &= 3(R + r). \end{aligned}$$

$\square$

3.35. $2r \leq \frac{AG + BG + CG}{3} \leq R$.

PROOF. We use the following facts:

- $\frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} = \frac{1}{r}$,
- $AG^2 + BG^2 + CG^2 = 3(R^2 - OG^2) \leq 3R^2$,

For the left-side inequality, note that the distances from G to BC, CA, AB are $\frac{h_a}{3}, \frac{h_b}{3}, \frac{h_c}{3}$, respectively, and so by the Erdős-Mordell Inequality and the Cauchy-Schwarz Inequality, we have

$$\begin{aligned}\frac{AG + BG + CG}{3} &\geq \frac{1}{3} \cdot 2 \left(\frac{h_a}{3} + \frac{h_b}{3} + \frac{h_c}{3} \right) \\ &= \frac{2}{9} \cdot r(h_a + h_b + h_c) \left(\frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} \right) \\ &\geq \frac{2}{9} \cdot r(1 + 1 + 1)^2 \\ &= 2r\end{aligned}$$

For the right-side inequality, we use the Cauchy-Schwarz Inequality, and get

$$\frac{AG + BG + CG}{3} \leq \frac{1}{3} \sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{AG^2 + BG^2 + CG^2} \leq \frac{1}{3} \sqrt{3} \cdot \sqrt{3R^2} = R.$$

□

3.36. $2r \leq \frac{AI + BI + CI}{3} \leq R.$

PROOF. For the left-side inequality, we have, by the Erdős-Mordell Inequality,

$$\frac{AI + BI + CI}{3} \geq \frac{1}{3}(2(r + r + r)) = 2r.$$

For the right-side inequality, let internal angle bisectors meet the circumcircle (of $\triangle ABC$) again at A', B', C' , respectively, and let AA', BB', CC' intersect the sides of $\triangle A'B'C'$ at D, E, F , respectively. Then $\triangle AB'C' \cong \triangle IB'C'$. Hence $B'C' \perp AI$ and $AD = ID$. Similarly, $C'A \perp BI$ and $EI = BE$, and $A'B' \perp CI$ and $FI = CF$.

Therefore, by the Erdős-Mordell Inequality,

$$\begin{aligned}\frac{AI + BI + CI}{3} &= \frac{1}{6} (AI + BI + CI + (AI + BI + CI)) \\ &= \frac{1}{6} (AI + BI + CI + 2(DI + EI + FI)) \\ &\leq \frac{1}{6} (AI + BI + CI + (A'I + B'I + C'I)) \\ &= \frac{1}{6} (AA' + BB' + CC') \\ &\leq \frac{1}{6} (2R + 2R + 2R) \\ &= R.\end{aligned}$$

□

3.37. $\frac{R_1}{a^2} + \frac{R_2}{b^2} + \frac{R_3}{c^2} \geq \frac{1}{R}.$

PROOF. By Mordell's Lemma and the AM-GM Inequality, we have

$$\begin{aligned}\frac{R_1}{a^2} + \frac{R_2}{b^2} + \frac{R_3}{c^2} &\geq \left(\frac{c}{a^3} r_2 + \frac{b}{a^3} r_3 \right) + \left(\frac{a}{b^3} r_3 + \frac{c}{b^3} r_1 \right) + \left(\frac{b}{c^3} r_1 + \frac{a}{c^3} r_2 \right) \\ &= \left(\frac{b}{c^3} + \frac{c}{b^3} \right) r_1 + \left(\frac{c}{a^3} + \frac{a}{c^3} \right) r_2 + \left(\frac{a}{b^3} + \frac{b}{a^3} \right) r_3 \\ &\geq 2r_1 \sqrt{\frac{b}{c^3} \cdot \frac{c}{b^3}} + 2r_2 \sqrt{\frac{c}{a^3} \cdot \frac{a}{c^3}} + 2r_3 \sqrt{\frac{a}{b^3} \cdot \frac{b}{a^3}} \\ &= \frac{2r_1}{bc} + \frac{2r_2}{ca} + \frac{2r_3}{ab} \\ &= \frac{2}{abc} (ar_1 + br_2 + cr_3) \\ &= \frac{4\Delta}{abc} \\ &= \frac{1}{R}.\end{aligned}$$

□

3.38. $\frac{m_a}{a} + \frac{m_b}{b} + \frac{m_c}{c} \geq \frac{s}{R}.$

PROOF. By the Erdős-Mordell Inequality (applied to the symmedian point K), we have

$$\begin{aligned} \frac{m_a}{a} + \frac{m_b}{b} + \frac{m_c}{c} &= \frac{a^2 + b^2 + c^2}{2abc} \left(\frac{2bcm_a}{a^2 + b^2 + c^2} + \frac{2cam_b}{a^2 + b^2 + c^2} + \frac{2abm_c}{a^2 + b^2 + c^2} \right) \\ &\geq \frac{a^2 + b^2 + c^2}{2abc} \cdot 2 \left(\frac{2a\Delta}{a^2 + b^2 + c^2} + \frac{2b\Delta}{a^2 + b^2 + c^2} + \frac{2c\Delta}{a^2 + b^2 + c^2} \right) \\ &= \frac{4s\Delta}{4R\Delta} \\ &= \frac{s}{R}, \end{aligned}$$

because, by the property of symmedians, we have

$$\frac{BK_a}{CK_a} = \frac{c^2}{b^2}$$

and so

$$BK_a = \frac{c^2}{b^2 + c^2} \cdot a \quad \text{and} \quad CK_a = \frac{b^2}{b^2 + c^2} \cdot a.$$

By Stewart's Theorem, we have

$$BK_a \cdot b^2 + CK_a \cdot c^2 = a (AK_a^2 + BK_a \cdot CK_a)$$

and so

$$AK_a^2 = \frac{2b^2c^2}{b^2 + c^2} - \frac{a^2b^2c^2}{(b^2 + c^2)^2} = \frac{4b^2c^2m_a^2}{(b^2 + c^2)^2}.$$

Since, by Menelaus's Theorem (applied to $\triangle ABK_a$ and the transversal K_cKC), we have

$$\frac{AK}{KK_a} = \frac{AK_c}{BK_c} \cdot \frac{BC}{CK_a} = \frac{b^2}{a^2} \cdot \frac{b^2 + c^2}{a^2} = \frac{b^2 + c^2}{a^2}.$$

Hence

$$AK = \frac{b^2 + c^2}{a^2 + b^2 + c^2} \cdot \frac{2bcm_a}{b^2 + c^2} = \frac{2bcm_a}{a^2 + b^2 + c^2}.$$

Similarly,

$$BK = \frac{2cam_b}{a^2 + b^2 + c^2} \quad \text{and} \quad CK = \frac{2abm_c}{a^2 + b^2 + c^2}.$$

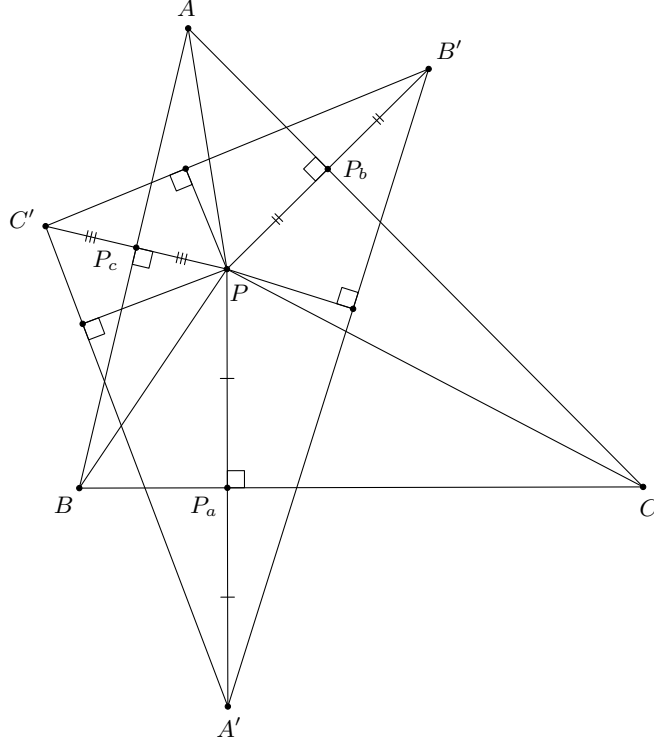
To compute the distances of K to the sides, it is enough to note that the ratio of the distances is given by $a : b : c$, by the property of symmedians. Therefore, by considering the area of $\triangle ABC$ with respect to these distances, the distances from K are given by

$$\frac{2a\Delta}{a^2 + b^2 + c^2}, \quad \frac{2b\Delta}{a^2 + b^2 + c^2} \quad \text{and} \quad \frac{2c\Delta}{a^2 + b^2 + c^2}.$$

□

$$\mathbf{3.39.} \quad \frac{1}{r_1 r_2} + \frac{1}{r_2 r_3} + \frac{1}{r_3 r_1} \geq 2 \left(\frac{1}{R_1 r_1} + \frac{1}{R_2 r_2} + \frac{1}{R_3 r_3} \right) \geq 4 \left(\frac{1}{R_1 R_2} + \frac{1}{R_2 R_3} + \frac{1}{R_3 R_1} \right).$$

PROOF.



Reflect P about BC, CA, AB to get the points A', B', C' , respectively. Consider $\triangle A'B'C'$ and its interior point P . Let R'_1, R'_2, R'_3 be the distances from P to A', B', C' , respectively, and let r'_1, r'_2, r'_3 be the distances from P to $B'C', C'A', A'B'$, respectively. Then, by considering the area of $\triangle A'PB'$ in two ways, we have

$$r'_3 = \frac{2[A'PB']}{A'B'} = \frac{\frac{1}{2}PA' \cdot PB' \sin \angle A'PB'}{2P_a P_b} = \frac{\frac{1}{2}(2r_1)(2r_2) \sin(180^\circ - C)}{2R_3 \sin C} = \frac{2r_1 r_2}{R_3}.$$

Similarly, we have

$$r'_1 = \frac{2r_2 r_3}{R_1} \quad \text{and} \quad r'_2 = \frac{2r_3 r_1}{R_2}.$$

By the Erdős-Mordell Inequality,

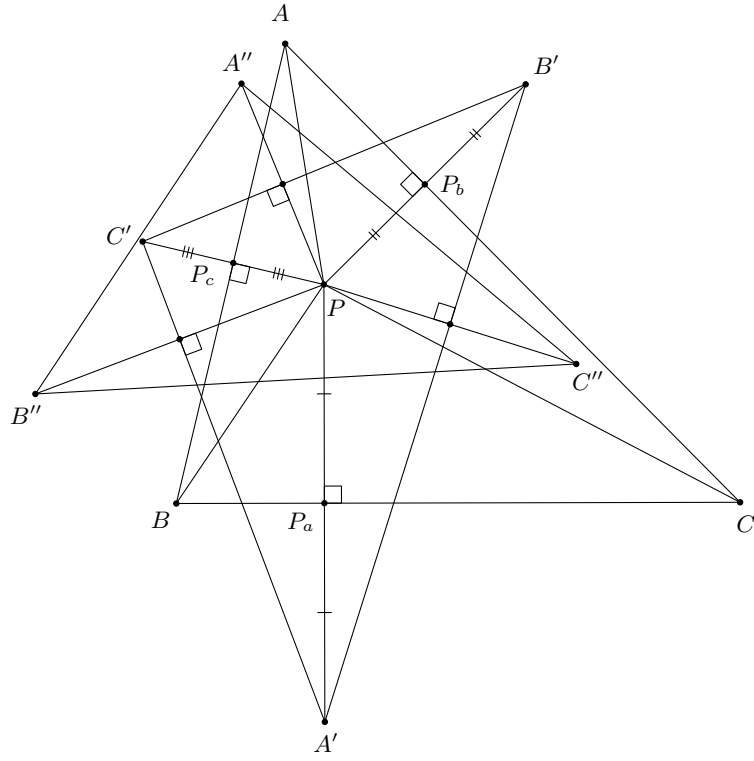
$$\begin{aligned} \frac{1}{r_1 r_2} + \frac{1}{r_2 r_3} + \frac{1}{r_3 r_1} &= \frac{1}{2r_1 r_2 r_3} (R'_1 + R'_2 + R'_3) \\ &\geq \frac{1}{2r_1 r_2 r_3} \cdot 2(r'_1 + r'_2 + r'_3) \\ &= 2 \left(\frac{1}{R_1 r_1} + \frac{1}{R_2 r_2} + \frac{1}{R_3 r_3} \right). \end{aligned}$$

This proves the left-side inequality. For the right-side inequality, we repeat the same process to P and $\triangle A'B'C'$ to obtain A'', B'', C'' . Then defining R''_1, R''_2, R''_3 as the distances from P to A'', B'', C'' , respectively, and r''_1, r''_2, r''_3 as the distances from P to $B''C'', C''A'', A''B''$, respectively, we get in a similar manner,

$$r''_3 = \frac{2r'_1 r'_2}{R'_3} = \frac{4r_1 r_2 r_3}{R_1 R_2}.$$

Similarly, we have

$$r''_1 = \frac{4r_1 r_2 r_3}{R_2 R_3} \quad \text{and} \quad r''_2 = \frac{4r_1 r_2 r_3}{R_3 R_1}.$$



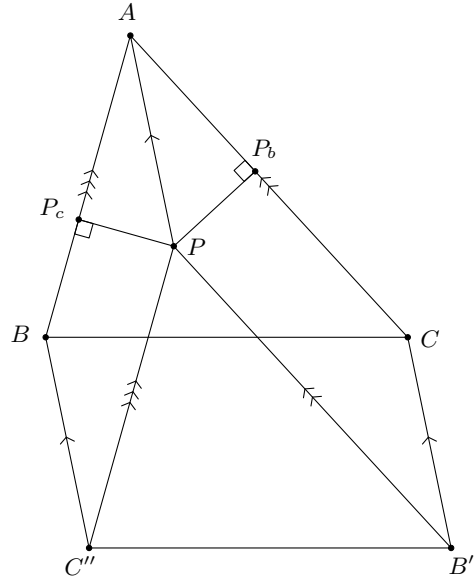
Therefore, by the Erdős-Mordell Inequality,

$$\begin{aligned}
2\left(\frac{1}{R_1 r_1} + \frac{1}{R_2 r_2} + \frac{1}{R_3 r_3}\right) &= \frac{1}{2r_1 r_2 r_3} \cdot 4\left(\frac{r_2 r_3}{R_1} + \frac{r_3 r_1}{R_2} + \frac{r_1 r_2}{R_3}\right) \\
&= \frac{1}{2r_1 r_2 r_3} \cdot (2r'_1 + 2r'_2 + 2r'_3) \\
&= \frac{1}{2r_1 r_2 r_3} \cdot (R''_1 + R''_2 + R''_3) \\
&\geq \frac{1}{2r_1 r_2 r_3} \cdot 2(r''_1 + r''_2 + r''_3) \\
&= \frac{1}{r_1 r_2 r_3} \cdot 4r_1 r_2 r_3 \left(\frac{1}{R_1 R_2} + \frac{1}{R_2 R_3} + \frac{1}{R_3 R_1}\right) \\
&= 4\left(\frac{1}{R_1 R_2} + \frac{1}{R_2 R_3} + \frac{1}{R_3 R_1}\right).
\end{aligned}$$

□

3.40. $R_1r_1 + R_2r_2 + R_3r_3 \geq 2(r_1r_2 + r_2r_3 + r_3r_1)$.

PROOF.



Let C'' , B'' be points on the same side as P with respect to AB, CA , respectively, such that $ABC''P$ and $APB''C$ are parallelograms. (This mimics the third proof of Mordell's Lemma where $C' = B$ and $B' = C$.) Then $BC''B''B$ is a parallelogram and

$$\begin{aligned}
 br_2 + cr_3 &= AC \cdot PP_b + AB \cdot PP_c \\
 &= [APB''C] + [ABC''P] \\
 &= [BC''B''C] \\
 &\leq BC \cdot B''C \\
 &= BC \cdot PA \\
 &= a \cdot R_1.
 \end{aligned}$$

Similarly, we have

$$bR_2 \geq cr_3 + ar_2 \quad \text{and} \quad cR_3 \geq ar_1 + br_2.$$

Therefore we rewrite and get

$$\begin{aligned}
 R_1r_1 + R_2r_2 + R_3r_3 &\geq \left(\frac{b}{a}r_2 + \frac{c}{a}r_3\right)r_1 + \left(\frac{c}{b}r_3 + \frac{a}{b}r_1\right)r_2 + \left(\frac{a}{c}r_1 + \frac{b}{c}r_2\right)r_3 \\
 &= \left(\frac{a}{b} + \frac{b}{a}\right)r_1r_2 + \left(\frac{b}{c} + \frac{c}{b}\right)r_2r_3 + \left(\frac{c}{a} + \frac{a}{c}\right)r_3r_1 \\
 &\geq 2r_1r_2\sqrt{\frac{a}{b} \cdot \frac{b}{a}} + 2r_2r_3\sqrt{\frac{b}{c} \cdot \frac{c}{b}} + 2r_3r_1\sqrt{\frac{c}{a} \cdot \frac{a}{c}} \\
 &= 2(r_1r_2 + r_2r_3 + r_3r_1).
 \end{aligned}$$

□

3.4 Miscellaneous Geometric Inequalities

3.41 (Hayashi's Inequality). $\frac{R_1 R_2}{ab} + \frac{R_2 R_3}{bc} + \frac{R_3 R_1}{ca} \geq 1$.

PROOF. We use complex numbers and we let z_1, z_2, z_3, z to be complex numbers representing the points A, B, C, P , respectively. Now we consider the quadratic polynomial in z given by

$$p(z) = \frac{(z - z_1)(z - z_2)}{(z_3 - z_1)(z_3 - z_2)} + \frac{(z - z_2)(z - z_3)}{(z_1 - z_2)(z_1 - z_3)} + \frac{(z - z_3)(z - z_1)}{(z_2 - z_3)(z_2 - z_1)}.$$

Since $p(z_1) = p(z_2) = p(z_3) = 1$, we have $p(z) \equiv 1$. Therefore, by the Triangle Inequality,

$$1 = |p(z)| \leq \left| \frac{(z - z_1)(z - z_2)}{(z_3 - z_1)(z_3 - z_2)} \right| + \left| \frac{(z - z_2)(z - z_3)}{(z_1 - z_2)(z_1 - z_3)} \right| + \left| \frac{(z - z_3)(z - z_1)}{(z_2 - z_3)(z_2 - z_1)} \right| = \frac{R_1 R_2}{ab} + \frac{R_2 R_3}{bc} + \frac{R_3 R_1}{ca}.$$

□

3.42. $\frac{m_a m_b}{ab} + \frac{m_b m_c}{bc} + \frac{m_c m_a}{ca} \geq \frac{9}{4}$.

PROOF. By Hayashi's Inequality,

$$\begin{aligned} \frac{m_a m_b}{ab} + \frac{m_b m_c}{bc} + \frac{m_c m_a}{ca} &= \frac{\frac{3}{2}AG \cdot \frac{3}{2}BG}{ab} + \frac{\frac{3}{2}BG \cdot \frac{3}{2}CG}{bc} + \frac{\frac{3}{2}CG \cdot \frac{3}{2}AG}{ca} \\ &= \frac{9}{4} \left(\frac{AG \cdot BG}{ab} + \frac{BG \cdot CG}{bc} + \frac{CG \cdot AG}{ca} \right) \\ &\geq \frac{9}{4} \cdot 1 \\ &= \frac{9}{4}. \end{aligned}$$

□

3.43. $m_a m_b c^2 + m_b m_c a^2 + m_c m_a b^2 \geq \frac{(a^2 + b^2 + c^2)^2}{4}$.

PROOF. We apply Hayashi's Inequality to the symmedian point K and get

$$\frac{AK \cdot BK}{ab} + \frac{BK \cdot CK}{bc} + \frac{CK \cdot AK}{ca} \geq 1$$

that is,

$$\frac{1}{(a^2 + b^2 + c^2)^2} \left(\frac{2bcm_a \cdot 2cam_b}{ab} + \frac{2cam_b \cdot 2abm_c}{bc} + \frac{2abm_c \cdot 2bcm_a}{ca} \right) \geq 1.$$

The results follow after rearranging.

□

3.44. $k_a(r + r_a) + k_b(r + r_b) + k_c(r + r_c) \leq 4(R + r)^2$.

PROOF. We use the following facts:

- $k_a = \frac{2bcm_a}{b^2 + c^2}, k_b = \frac{2cam_b}{c^2 + a^2}, k_c = \frac{2abm_c}{a^2 + b^2}$
- $k_a \leq m_a \leq R(1 + \cos A), k_b \leq m_b \leq R(1 + \cos B), k_c \leq m_c \leq R(1 + \cos C)$ (by the AM-GM Inequality and the Triangle Inequality)
- $r_a(s - a) = r_b(s - b) = r_c(s - c) = rs = \Delta$.

By Gerretsen's Inequality,

$$\begin{aligned} k_a(r + r_a) + k_b(r + r_b) + k_c(r + r_c) &\leq R(1 + \cos A) \left(\frac{\Delta}{s} + \frac{\Delta}{s - a} \right) + R(1 + \cos B) \left(\frac{\Delta}{s} + \frac{\Delta}{s - b} \right) + R(1 + \cos C) \left(\frac{\Delta}{s} + \frac{\Delta}{s - c} \right) \\ &= 2Rrs \left(\frac{s(s - a)}{bc} \cdot \frac{b + c}{s(s - a)} + \frac{s(s - b)}{ca} \cdot \frac{c + a}{s(s - b)} + \frac{s(s - c)}{ab} \cdot \frac{a + b}{s(s - c)} \right) \\ &= 2Rrs \left(\frac{a(b + c)}{abc} + \frac{b(c + a)}{abc} + \frac{c(a + b)}{abc} \right) \\ &= \frac{4Rrs}{4Rrs} (ab + bc + ca) \\ &= s^2 + 4Rr + r^2 \\ &\leq (4R^2 + 4Rr + 3r^2) + 4Rr + r^2 \\ &= 4(R + r)^2. \end{aligned}$$

□

3.45. $\frac{w_a}{v_a \sin^2 A} + \frac{w_b}{v_b \sin^2 B} + \frac{w_c}{v_c \sin^2 C} \geq 3.$

PROOF. Observe that

- $h_a = b \sin C$, $h_b = c \sin A$ and $h_c = a \sin B$, so that $h_a h_b h_c = abc \sin A \sin B \sin C$;
- $w_a \geq h_a$, $w_b \geq h_b$, $w_c \geq h_c$;
- $2R \geq v_a, v_b, v_c$.

Hence, using the AM-GM Inequality, we have

$$\begin{aligned} \frac{w_a}{v_a \sin^2 A} + \frac{w_b}{v_b \sin^2 B} + \frac{w_c}{v_c \sin^2 C} &\geq \frac{h_a}{2R \sin^2 A} + \frac{h_b}{2R \sin^2 B} + \frac{h_c}{2R \sin^2 C} \\ &= \frac{h_a}{a \sin A} + \frac{h_b}{b \sin B} + \frac{h_c}{c \sin C} \\ &\geq 3 \sqrt[3]{\frac{h_a h_b h_c}{abc \sin A \sin B \sin C}} \\ &= 3. \end{aligned}$$

□

3.46. $\frac{R_1}{h_b + h_c} + \frac{R_2}{h_c + h_a} + \frac{R_3}{h_a + h_b} \geq 1.$

PROOF. By Mordell's Lemma and by again mimicking its third proof, we have

$$aR_1 \geq br_3 + cr_2 \quad \text{and} \quad aR_1 \geq br_2 + cr_3.$$

We add and get

$$R_1 \geq \frac{(b+c)(r_2+r_3)}{2a}.$$

Similarly, we have

$$R_2 \geq \frac{(c+a)(r_3+r_1)}{2b} \quad \text{and} \quad R_3 \geq \frac{(a+b)(r_1+r_2)}{2c}.$$

Therefore, together with the AM-GM Inequality, we have

$$\begin{aligned} \frac{R_1}{h_b + h_c} + \frac{R_2}{h_c + h_a} + \frac{R_3}{h_a + h_b} &= \frac{1}{2\Delta} \left(\frac{bcR_1}{b+c} + \frac{caR_2}{c+a} + \frac{abR_3}{a+b} \right) \\ &\geq \frac{1}{2\Delta} \left(\frac{bc}{b+c} \cdot \frac{(b+c)(r_2+r_3)}{2a} + \frac{ca}{c+a} \cdot \frac{(c+a)(r_3+r_1)}{2b} + \frac{ab}{a+b} \cdot \frac{(a+b)(r_1+r_2)}{2c} \right) \\ &= \frac{1}{4\Delta} \left(\frac{bc(r_2+r_3)}{a} + \frac{ca(r_3+r_1)}{b} + \frac{ab(r_1+r_2)}{c} \right) \\ &= \frac{1}{4abc\Delta} (b^2c^2(r_2+r_3) + c^2a^2(r_3+r_1) + a^2b^2(r_1+r_2)) \\ &= \frac{1}{4abc\Delta} ((a^2b^2 + c^2a^2)r_1 + (b^2c^2 + a^2b^2)r_2 + (c^2a^2 + b^2c^2)r_3) \\ &\geq \frac{1}{4abc\Delta} (2a^2bcr_1 + 2ab^2cr_2 + 2abc^2r_3) \\ &= \frac{1}{2\Delta} (ar_1 + br_2 + cr_3) \\ &= 1. \end{aligned}$$

□

3.47. Extend AG, BG, CG to meet the circumcircle of $\triangle ABC$ at A', B', C' respectively. Then

$$AG + BG + CG \leq GA' + GB' + GC'.$$

PROOF. By the Power of a Point Theorem,

$$m_a \cdot M_a A' = M_a A \cdot M_a A' = M_a B \cdot M_a C = \frac{a^2}{4},$$

and so

$$M_a A' = \frac{a^2}{4m_a}.$$

This implies that

$$GA' = GM_a + M_a A' = \frac{m_a}{3} + \frac{a^2}{4m_a} = \frac{a^2 + b^2 + c^2}{6m_a}.$$

Similarly,

$$GB' = \frac{a^2 + b^2 + c^2}{6m_b} \quad \text{and} \quad GC' = \frac{a^2 + b^2 + c^2}{6m_c}.$$

Now, using the fact that $a^2 + b^2 + c^2 = \frac{4}{3}(m_a^2 + m_b^2 + m_c^2)$, we have

$$\begin{aligned}
 GA' + GB' + GC' &= \left(\frac{a^2 + b^2 + c^2}{6} \right) \left(\frac{1}{m_a} + \frac{1}{m_b} + \frac{1}{m_c} \right) \\
 &= \frac{2}{9} (m_a^2 + m_b^2 + m_c^2) \left(\frac{1}{m_a} + \frac{1}{m_b} + \frac{1}{m_c} \right) \\
 &= \frac{2}{3} \cdot \frac{1}{3} (m_a^2 + m_b^2 + m_c^2) \left(\frac{1}{m_a} + \frac{1}{m_b} + \frac{1}{m_c} \right) \\
 &\geq \frac{2}{3} (m_a + m_b + m_c) \\
 &= AG + BG + CG,
 \end{aligned}$$

by Chebyshev's Inequality. □

3.48. $\frac{2}{3}(\cos A + \cos B + \cos C) + \frac{a^3 + b^3 + c^3}{3abc} \leq \frac{R}{r}.$

PROOF. We use the following facts:

- $\cos A + \cos B + \cos C = 1 + \frac{r}{R},$
- $a^3 + b^3 + c^3 = 3abc + (a + b + c)(a^2 + b^2 + c^2 - (ab + bc + ca)) = 2s(s^2 - 3r^2 - 6Rr),$
- $abc = 4Rrs,$

and rewrite the desired inequality as

$$\frac{2}{3} \left(1 + \frac{r}{R} \right) + \frac{2s(s^2 - 3r^2 - 6Rr)}{12Rrs} \leq \frac{R}{r}$$

or equivalently,

$$s^2 \leq 6R^2 + 2Rr - r^2.$$

By Gerretsen's Inequality, it is enough to show that

$$4R^2 + 4Rr + 3r^2 \leq 6R^2 + 2Rr - r^2.$$

Now, by Euler's Inequality,

$$6R^2 + 2Rr - r^2 - (4R^2 + 4Rr + 3r^2) = 2R^2 - 2Rr - 4r^2 = 2(R - 2r)(R + r) \geq 0.$$

□

3.49. If $\triangle ABC$ is acute-angled, then $9 \left(\frac{AG^2 + BG^2 + CG^2}{AO^2 + BO^2 + CO^2} \right) - \frac{AH \cdot BH \cdot CH}{AI \cdot BI \cdot CI} \leq 8.$

PROOF. We use the following facts:

- $AG^2 + BG^2 + CG^2 = 3(R^2 - OG^2),$
- $AH \cdot BH \cdot CH = 8R^3 \cos A \cos B \cos C,$
- $AI \cdot BI \cdot CI = 64R^3 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2},$
- $9OG^2 = (3OG)^2 = OH^2 = R^2 + 4R^2 \cos A(\cos A - \sin B \sin C) - 4R^2 \cos A \cos B \cos C = R^2(1 - 8 \cos A \cos B \cos C),$
- $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}.$

Then

$$\begin{aligned}
 9 \left(\frac{AG^2 + BG^2 + CG^2}{AO^2 + BO^2 + CO^2} \right) - \frac{AH \cdot BH \cdot CH}{AI \cdot BI \cdot CI} &= 9 \left(\frac{3(R^2 - OG^2)}{3R^2} \right) - \frac{8R^3 \cos A \cos B \cos C}{64R^3 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2}} \\
 &= \frac{9R^2 - 9OG^2}{R^2} - \frac{\cos A \cos B \cos C}{8 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2}} \\
 &= \frac{9R^2 - R^2(1 - 8 \cos A \cos B \cos C)}{R^2} - \frac{\cos A \cos B \cos C}{8 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2}} \\
 &= 8 + 8 \cos A \cos B \cos C - \frac{\cos A \cos B \cos C}{8 \sin^2 \frac{A}{2} \sin^2 \frac{B}{2} \sin^2 \frac{C}{2}} \\
 &\leq 8 + 8 \cos A \cos B \cos C - \frac{\cos A \cos B \cos C}{8 \cdot \frac{1}{64}} \\
 &= 8.
 \end{aligned}$$

□

$$3.50. \frac{1}{AI \cdot BI} + \frac{1}{BI \cdot CI} + \frac{1}{CI \cdot AI} \leq \frac{R+r}{2Rr^2}.$$

PROOF. We use the following facts:

- $AI = 4R \sin \frac{B}{2} \sin \frac{C}{2}$, $BI = 4R \sin \frac{C}{2} \sin \frac{A}{2}$, $CI = 4R \sin \frac{A}{2} \sin \frac{B}{2}$,
- $r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$,
- $\frac{AI \cdot BI \cdot CI}{2Rr^2} = 2$,
- $ab + bc + ca = r^2 + s^2 + 4Rr$,
- $AI = \frac{bc}{s} \cos \frac{A}{2} = \frac{\sqrt{bc}\sqrt{s(s-a)}}{s}$, $BI = \frac{ca}{s} \cos \frac{B}{2} = \frac{\sqrt{ca}\sqrt{s(s-b)}}{s}$, $CI = \frac{ab}{s} \cos \frac{C}{2} = \frac{\sqrt{ab}\sqrt{s(s-c)}}{s}$.

Observe that the desired inequality is equivalent to

$$AI + BI + CI \leq 2(R+r).$$

Now, by the Cauchy-Schwarz Inequality and Gerretsen's Inequality, we have

$$\begin{aligned} (AI + BI + CI)^2 &= \frac{1}{s^2} \left(\sqrt{bc}\sqrt{s(s-a)} + \sqrt{ca}\sqrt{s(s-b)} + \sqrt{ab}\sqrt{s(s-c)} \right)^2 \\ &\leq \frac{1}{s^2} (bc + ca + ab) (s(s-a) + s(s-b) + s(s-c)) \\ &= ab + bc + ca \\ &= r^2 + s^2 + 4Rr \\ &\leq r^2 + (4R^2 + 4Rr + 3r^2) + 4Rr \\ &= 4(R+r)^2. \end{aligned}$$

The result follows. □

$$3.51. \text{ If } \triangle ABC \text{ is acute-angled, then } \sqrt{AN} + \sqrt{BN} + \sqrt{CN} \leq \sqrt{\frac{15R+6r}{2}}.$$

PROOF. Since AN is a median of $\triangle AHO$, we have

$$AN \leq \frac{1}{2}(AO + AH) = \frac{1}{2}(R + 2R \cos A).$$

Similarly, we have

$$BN \leq \frac{1}{2}(R + 2R \cos B) \quad \text{and} \quad CN \leq \frac{1}{2}(R + 2R \cos C).$$

By the Cauchy-Schwarz Inequality, we have

$$\begin{aligned} \sqrt{AN} + \sqrt{BN} + \sqrt{CN} &\leq \sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{AN + BN + CN} \\ &= \sqrt{\frac{3}{2}} (3R + 2R(\cos A + \cos B \cos C)) \\ &= \sqrt{\frac{3}{2}} \left(3R + 2R \left(1 + \frac{r}{R} \right) \right) \\ &= \sqrt{\frac{15R+6r}{2}}. \end{aligned}$$

□

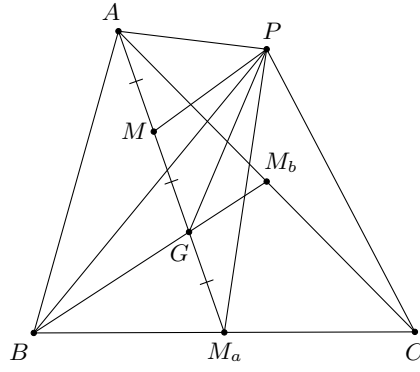
$$3.52. AN^2 + BN^2 + CN^2 \geq 12r^2.$$

PROOF. We use the following facts:

- $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca) = 4s^2 - 2(r^2 + s^2 + 4Rr) = 2s^2 - 2r^2 - 8Rr$,
- $OH^2 = 9R^2 - (a^2 + b^2 + c^2)$.

We also use the fact that for any point P (on the plane of $\triangle ABC$), we have

$$AP^2 + BP^2 + CP^2 = AG^2 + BG^2 + CG^2 + 3PG^2.$$



Indeed, let M be the midpoint of AG . We apply the formula for the median to $\triangle BCP$, $\triangle AGP$ and $\triangle MM_aP$ and get

$$\begin{aligned} BP^2 + CP^2 &= 2MM_a^2 + \frac{1}{2}a^2, \\ MP^2 + M_aP^2 &= 2PG^2 + \frac{1}{2}MM_a^2, \\ AP^2 + PG^2 &= 2MP^2 + \frac{1}{2}AG^2. \end{aligned}$$

We multiply the second equality by 2 and add them to obtain

$$AP^2 + BP^2 + CP^2 - 3PG^2 = \frac{1}{2}a^2 + \frac{3}{2}AG^2.$$

Hence

$$3(AP^2 + BP^2 + CP^2 - 3PG^2) = \frac{1}{2}(a^2 + b^2 + c^2) + \frac{3}{2}(AG^2 + BG^2 + CG^2) = 3(AG^2 + BG^2 + CG^2).$$

To prove the desired inequality, we choose $P = N$ and get

$$\begin{aligned} AN^2 + BN^2 + CN^2 &= AG^2 + BG^2 + CG^2 + 3NG^2 \\ &= \frac{4}{9} \cdot \frac{3}{4}(a^2 + b^2 + c^2) + 3\left(\frac{1}{6}OH\right)^2 \\ &= \frac{1}{12}(9R^2 - (a^2 + b^2 + c^2) + 4(a^2 + b^2 + c^2)) \\ &= \frac{1}{4}(3R^2 + 2s^2 - 2r^2 - 8Rr) \\ &\geq \frac{1}{4}(3R^2 + 2(16Rr - 5r^2) - 2r^2 - 8Rr) \\ &= \frac{1}{4}(3R^2 + 24Rr - 12r^2) \\ &\geq \frac{1}{4}(3(2r)^2 + 24(2r)r - 12r^2) \\ &= 12r^2, \end{aligned}$$

by Gerretsen's Inequality and Euler's Inequality. □

3.53. $\frac{AG_e \cdot AN_a}{G_e G_a \cdot N_a N_a} + \frac{BG_e \cdot BN_a}{G_e G_b \cdot N_a N_b} + \frac{CG_e \cdot CN_a}{G_e G_c \cdot N_a N_c} \geq 12.$

PROOF. We use the following facts:

- $ab + bc + ca = r^2 + s^2 + 4Rr,$
- $a^3 + b^3 + c^3 = 3abc + (a + b + c)(a^2 + b^2 + c^2 - (ab + bc + ca)) = 2s(s^2 - 3r^2 - 6Rr),$

By van Aubel's Theorem,

$$\frac{AG_e}{G_e G_a} = \frac{s-a}{s-b} + \frac{s-a}{s-c} = (s-a) \cdot \frac{2s-(b+c)}{(s-b)(s-c)} = \frac{a(s-a)}{(s-b)(s-c)}$$

and

$$\frac{AN_a}{N_a N_a} = \frac{s-b}{s-a} + \frac{s-c}{s-a} = \frac{2s-(b+c)}{s-a} = \frac{a}{s-a}.$$

Similarly, we have

$$\frac{BG_e}{G_e G_b} = \frac{b(s-b)}{(s-c)(s-a)}, \quad \frac{CG_e}{G_e G_c} = \frac{c(s-c)}{(s-a)(s-b)}, \quad \frac{BN_a}{N_a N_b} = \frac{b}{s-b}, \quad \text{and} \quad \frac{CN_a}{N_a N_c} = \frac{c}{s-c}.$$

Then, by Euler's Inequality,

$$\begin{aligned}
\frac{AG_e \cdot AN_a}{G_e G_a \cdot N_a N_a} + \frac{BG_e \cdot BN_a}{G_e G_b \cdot N_a N_b} + \frac{CG_e \cdot CN_a}{G_e G_c \cdot N_a N_c} &= \frac{a^2(s-a)}{(s-a)(s-b)(s-c)} + \frac{b^2(s-b)}{(s-a)(s-b)(s-c)} + \frac{c^2(s-c)}{(s-a)(s-b)(s-c)} \\
&= \frac{(a^2 + b^2 + c^2)s - (a^3 + b^3 + c^3)}{r^2 s} \\
&= \frac{2s(s^2 - r^2 - 4Rr) - 2s(s^2 - 3r^2 - 6Rr)}{r^2 s} \\
&= \frac{4R}{r} + 4 \\
&\geq 4 \cdot 2 + 4 \\
&= 12.
\end{aligned}$$

□

3.54. $\sin A \sin B \sin C \left(\frac{1}{\sin A + \sin B} + \frac{1}{\sin B + \sin C} + \frac{1}{\sin C + \sin A} \right) \leq \frac{3}{4}(\cos A + \cos B + \cos C).$

PROOF. We use the following facts:

- $\sin^2 A + \sin^2 B + \sin^2 C = \frac{a^2}{4R^2} + \frac{b^2}{4R^2} + \frac{c^2}{4R^2} = \frac{s^2 - 4Rr - r^2}{2R^2},$
- $\sin A \sin B + \sin B \sin C + \sin C \sin A = \frac{ab}{4R^2} + \frac{bc}{4R^2} + \frac{ca}{4R^2} = \frac{r^2 + s^2 + 4Rr}{4R^2},$
- $\cos A + \cos B + \cos C = 1 + \frac{r}{R}.$

Let L be the left-hand side of the inequality. Then

$$\begin{aligned}
L &\leq \sin A \sin B \sin C \left(\frac{1}{2\sqrt{\sin A \sin B}} + \frac{1}{2\sqrt{\sin B \sin C}} + \frac{1}{2\sqrt{\sin C \sin A}} \right) \\
&= \frac{1}{2} \left(\sin A \sqrt{\sin B \sin C} + \sin B \sqrt{\sin C \sin A} + \sin C \sqrt{\sin A \sin B} \right) \\
&\leq \frac{1}{2} \sqrt{\sin^2 A + \sin^2 B + \sin^2 C} \cdot \sqrt{\sin B \sin C + \sin C \sin A + \sin A \sin B} \\
&\leq \frac{1}{2} \sqrt{\frac{(4R^2 + 4Rr + 3r^2) - 4Rr - r^2}{2R^2}} \cdot \sqrt{\frac{r^2 + (4R^2 + 4Rr + 3r^2) + 4Rr}{4R^2}} \\
&= \frac{1}{2} \sqrt{\frac{4R^2 + 2r^2}{2R^2}} \cdot \sqrt{\frac{4R^2 + 8Rr + 4r^2}{4R^2}} \\
&= \frac{1}{2} \sqrt{2 + \left(\frac{r}{R}\right)^2} \cdot \left(1 + \frac{r}{R}\right) \\
&\leq \frac{1}{2} \sqrt{2 + \left(\frac{1}{2}\right)^2} (\cos A + \cos B + \cos C) \\
&= \frac{3}{4}(\cos A + \cos B + \cos C),
\end{aligned}$$

where we used the AM-GM Inequality in the first inequality, Cauchy-Schwarz Inequality in the second inequality, Gerretsen's Inequality in the third inequality, and finally, Euler's Inequality in the last inequality. □

3.55. $a^2 \tan \frac{A}{2} + b^2 \tan \frac{B}{2} + c^2 \tan \frac{C}{2} \leq 6\sqrt{3}R(R-r).$

PROOF. We use the following facts:

- $\tan \frac{A}{2} = \frac{r}{s-a}, \tan \frac{B}{2} = \frac{r}{s-b}, \tan \frac{C}{2} = \frac{r}{s-c},$
- $\frac{a}{s-a} + \frac{b}{s-b} + \frac{c}{s-c} = s \left(\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} \right) - 3 = \frac{4R-2r}{r},$
- $s = \frac{a+b+c}{2} \leq \frac{3\sqrt{3}R}{2}.$

Therefore

$$\begin{aligned}
a^2 \tan \frac{A}{2} + b^2 \tan \frac{B}{2} + c^2 \tan \frac{C}{2} &= \frac{a^2 r}{s-a} + \frac{b^2 r}{s-b} + \frac{c^2 r}{s-c} \\
&= r \left(\frac{a^2}{s-a} + a + \frac{b^2 r}{s-b} + b + \frac{c^2 r}{s-c} + c \right) - 2rs \\
&= rs \left(\frac{a}{s-a} + \frac{b}{s-b} + \frac{c}{s-c} \right) - 2rs \\
&= rs \left(\frac{4R-2r}{r} \right) - 2rs \\
&= 4(R-r)s \\
&\leq 4(R-r) \cdot \frac{3\sqrt{3}R}{2} \\
&= 6\sqrt{3}R(R-r).
\end{aligned}$$

□

3.56. $\frac{r_a^2}{h_a} + \frac{r_b^2}{h_b} + \frac{r_c^2}{h_c} \geq 3(2R-r).$

PROOF. We use the following facts:

- $r_a + r_b + r_c = 4R + r,$
- $r_a^2 + r_b^2 + r_c^2 = (r_a + r_b + r_c)^2 - 2(r_a r_b + r_b r_c + r_c r_a) = (4R + r)^2 - 2s^2,$
- $\frac{1}{r} - \frac{1}{r_a} = \frac{s}{\Delta} - \frac{s-a}{\Delta} = \frac{2}{h_a}, \quad \frac{1}{r} - \frac{1}{r_b} = \frac{2}{h_b}, \quad \frac{1}{r} - \frac{1}{r_c} = \frac{2}{h_c}.$

Then, by Gerretsen's Inequality and by Euler's Inequality, we have

$$\begin{aligned}
\frac{r_a^2}{h_a} + \frac{r_b^2}{h_b} + \frac{r_c^2}{h_c} &= \frac{1}{2} \left(r_a^2 \left(\frac{1}{r} - \frac{1}{r_a} \right) + r_b^2 \left(\frac{1}{r} - \frac{1}{r_b} \right) + r_c^2 \left(\frac{1}{r} - \frac{1}{r_c} \right) \right) \\
&= \frac{1}{2} \left(\frac{1}{r} (r_a^2 + r_b^2 + r_c^2) - (r_a + r_b + r_c) \right) \\
&= \frac{1}{2} \left(\frac{1}{r} \cdot ((4R+r)^2 - s^2) - (4R+r) \right) \\
&= \frac{1}{r} (8R^2 + 2Rr - s^2) \\
&\geq \frac{1}{r} (8R^2 + 2Rr - (4R^2 + 4Rr + 3r^2)) \\
&= \frac{1}{r} (4R^2 - 2Rr - 3r^2) \\
&= \frac{1}{r} (4R(R-2r) + 3r(2R-r)) \\
&\geq \frac{1}{r} (0 + 3r(2R-r)) \\
&= 3(2R-r).
\end{aligned}$$

3.57. $\frac{a^3}{r_a} + \frac{b^3}{r_b} + \frac{c^3}{r_c} \leq \frac{abc}{r}.$

PROOF. We use $r_a = \frac{\Delta}{s-a}, r_b = \frac{\Delta}{s-b}$ and $r_c = \frac{\Delta}{s-b},$ and rewrite the desired inequality as

$$\begin{aligned}
&a^3(s-a) + b^3(s-b) + c^3(s-c) \leq abcs \\
\iff &\left(\frac{a+b+c}{2} \right) (a^3 + b^3 + c^3) - (a^4 + b^4 + c^4) \leq abc \left(\frac{a+b+c}{2} \right) \\
\iff &a^2(a-b)(a-c) + b^2(b-c)(b-a) + c^2(c-a)(c-b) \geq 0,
\end{aligned}$$

which is true by Schur's Inequality.

□

3.58. $\sqrt[3]{\frac{r_a}{h_a}} + \sqrt[3]{\frac{r_b}{h_b}} + \sqrt[3]{\frac{r_c}{h_c}} \leq \frac{3R}{2r}.$

PROOF. We use the following facts: $r_a + r_b + r_c = 4R + r$ and $\frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} = \frac{1}{r}.$

Then, by Hölder's Inequality,

$$\begin{aligned} \sqrt[3]{\frac{r_a}{h_a}} + \sqrt[3]{\frac{r_b}{h_b}} + \sqrt[3]{\frac{r_c}{h_c}} &= 1 \cdot \sqrt[3]{r_a} \cdot \frac{1}{\sqrt[3]{h_a}} + 1 \cdot \sqrt[3]{r_b} \cdot \frac{1}{\sqrt[3]{h_b}} + 1 \cdot \sqrt[3]{r_c} \cdot \frac{1}{\sqrt[3]{h_c}} \\ &\leq \sqrt[3]{1+1+1} \cdot \sqrt[3]{r_a+r_b+r_c} \cdot \sqrt[3]{\frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c}} \\ &= \sqrt[3]{3(4R+r)} \cdot \frac{1}{r} \\ &= \sqrt[3]{3(4t+1)}, \end{aligned}$$

where $t = \frac{R}{r}.$

Hence it is enough to prove that $\sqrt[3]{3(4t+1)} \leq \frac{3t}{2}.$ Now, observe that

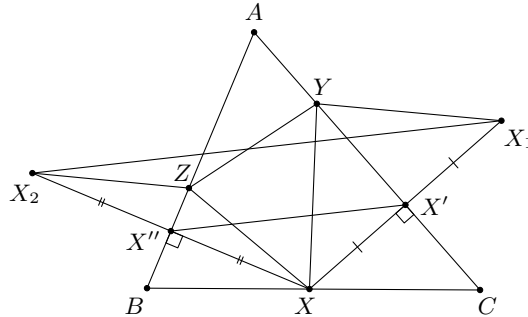
$$\begin{aligned} \sqrt[3]{3(4t+1)} \leq \frac{3t}{2} &\iff 8(4t+1) \leq 9t^3 \\ &\iff 9t^3 - 32t - 8 \geq 0 \\ &\iff (t-2)(9t^2 + 18t + 4) \geq 0, \end{aligned}$$

which is true since $t = \frac{R}{r} \geq 2$ by Euler's Inequality. □

3.59. Let X, Y, Z be points on the sides BC, CA, AB respectively of $\triangle ABC$. Then

$$(XY + YZ + ZX) \left(\frac{1}{AX} + \frac{1}{BY} + \frac{1}{CZ} \right) \geq \frac{a+b+c}{R}.$$

PROOF.



Let $\ell = \frac{XY + YZ + ZX}{2}.$ Reflect X about CA to get X_1 , with XX_1 intersecting CA at $X'.$ Reflect X about AB to get X_2 , with XX_2 intersecting AB at $X''.$ Then

$$2\ell = XY + YZ + ZX = X_1Y + YZ + ZX_2 \geq X_1X_2 = 2X'X'',$$

so $\ell \geq X'X''.$

Now observe that $AX'X''$ is cyclic with AX as diameter, so by the Sine Formula,

$$\frac{X'X''}{\sin A} = \frac{AX'}{\sin \angle AXX'} = AX.$$

Hence

$$\frac{\ell}{AX} \geq \frac{X'X''}{AX} = \sin A.$$

Similarly,

$$\frac{\ell}{BY} \geq \sin B \quad \text{and} \quad \frac{\ell}{CZ} \geq \sin C.$$

Therefore, add and get

$$\ell \left(\frac{1}{AX} + \frac{1}{BY} + \frac{1}{CZ} \right) \geq \sin A + \sin B + \sin C = \frac{a+b+c}{2R},$$

and the result follows. □

$$\mathbf{3.60.} \quad \frac{2bc \cos A}{b+c} < b+c-a < \frac{2bc}{a}$$

PROOF. If $a \leq \max\{b, c\}$, then $\max\{b, c\} < a + \min\{b, c\}$ and so

$$\begin{aligned} b+c-a &= \min\{b, c\} + \max\{b, c\} - a \\ &< \min\{b, c\} + \min\{b, c\} + a - a \\ &= 2 \min\{b, c\} \\ &\leq 2 \min\{b, c\} \cdot \frac{\max\{b, c\}}{a} \\ &= \frac{2bc}{a}. \end{aligned}$$

If $a > \max\{b, c\}$, then $a < 2 \max\{b, c\}$ must hold (otherwise $a \geq b+c$), and so

$$\begin{aligned} b+c-a &= \min\{b, c\} + \max\{b, c\} - a \\ &< \min\{b, c\} + a - a \\ &= \min\{b, c\} \\ &< \min\{b, c\} \cdot \frac{2 \max\{b, c\}}{a} \\ &= \frac{2bc}{a}. \end{aligned}$$

Hence right-side inequality holds.

Now observe that

$$(b+c)(b+c-a) - 2bc \cos A = a \left(\frac{2bc}{a} - (b+c-a) \right) > 0$$

which is true because of the right-side inequality. □

$$\mathbf{3.61.} \quad 2 \left(\cos \left(\frac{A-B}{2} \right) + \cos \left(\frac{B-C}{2} \right) + \cos \left(\frac{C-A}{2} \right) \right) \leq 3 + \sqrt{3 + 2(\cos(A-B) + \cos(B-C) + \cos(C-A))}$$

PROOF. Let $u = \cos A + i \sin A$, $v = \cos B + i \sin B$, and $w = \cos C + i \sin C$. Then $|u| = |v| = |w| = 1$ and

$$\begin{aligned} |u+v| &= |(\cos A + \cos B) + i(\sin A + \sin B)| \\ &= \left| 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) + 2i \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \right| \\ &= \left| 2 \cos \left(\frac{A+B}{2} \right) \left(\cos \left(\frac{A-B}{2} \right) + i \sin \left(\frac{A-B}{2} \right) \right) \right| \\ &= 2 \cos \left(\frac{A+B}{2} \right). \end{aligned}$$

Similarly,

$$|v+w| = 2 \cos \left(\frac{B+C}{2} \right) \quad \text{and} \quad |w+u| = 2 \cos \left(\frac{C+A}{2} \right).$$

Now, observe that

$$\begin{aligned} |u+v+w| &= \sqrt{(\cos A + \cos B + \cos C)^2 + (\sin A + \sin B + \sin C)^2} \\ &= \sqrt{3 + 2(\cos(A-B) + \cos(B-C) + \cos(C-A))}. \end{aligned}$$

By Hlawka's Inequality, we have

$$|u+v| + |v+w| + |w+u| \leq |u| + |v| + |w| + |u+v+w|,$$

and the result follows.

For completeness, we prove Hlawka's Inequality.

Observe that

$$|w+u||u+v| = |u(u+v+w) + vw| \leq |u(u+v+w)| + |vw|.$$

Similarly

$$|u+v||v+w| \leq |v(u+v+w)| + |uw| \quad \text{and} \quad |v+w||w+u| \leq |w(u+v+w)| + |uv|.$$

Then we add these inequalities, together with the fact that

$$|u+v|^2 + |v+w|^2 + |w+u|^2 = |u|^2 + |v|^2 + |w|^2 + |u+v+w|^2,$$

we get

$$(|u+v| + |v+w| + |w+u|)^2 \leq (|u| + |v| + |w| + |u+v+w|)^2.$$

□